
SPLIT-VLM: Salience-Guided Partitioning towards Local Coverage for Importance-Aware Token Dropping in Vision-Language Models

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Abstract

Large-scale vision–language models (VLMs) excel at multimodal reasoning, yet efficiency collapses when vision tokens—often orders of magnitude more than text—dominate compute and memory. Prior token-reduction strategies typically trade off salience (which is prone to position bias and incurs extra computation) against diversity (which can under-cover salient regions and is sensitive to hyperparameters). We present *SPLIT*, a theoretically grounded framework that jointly preserves salience and diversity while aggressively eliminating redundancy. *SPLIT* (i) estimates token importance via temporal shifts of hidden states across layers—eschewing attention scores and their biases; (ii) assigns adaptive region-level budgets to guarantee localized coverage; and (iii) selects tokens using a diversity score that prioritizes distinctive, non-redundant representations. Our analysis shows that adaptive budgeting yields tighter coverage guarantees than uniform allocation, and our selection rule maintains diversity without costly tuning. Empirically, *SPLIT* consistently outperforms state-of-the-art on image and video understanding benchmarks. On image understanding with LLaVA-1.5-7B, *SPLIT* preserves over 99% accuracy with 192 vision tokens and about 92.8% with only 64 tokens, demonstrating robust performance under severe token budgets. These results indicate that *SPLIT* delivers scalable, attention-score-free token reduction that makes multimodal reasoning substantially more efficient without sacrificing accuracy.

1. Introduction

Recent large-scale vision-language models (VLMs) have shown strong performance on tasks such as image captioning, visual question answering, and multimodal dialogue (Alayrac et al., 2022; Li et al., 2023a; Radford et al., 2021; Liu et al., 2023), establishing a new paradigm of AI that jointly understands vision and language. These advances are now expanding beyond benchmark improvements to real-world applications, including robotics and physical environments (Sapkota et al., 2025; Zhou et al., 2025b). However, a fundamental bottleneck remains: the excessive number of vision tokens generated from images. Transformer-based VLMs split an image into hundreds or thousands of patch tokens, which are processed together with text tokens. For instance, a 224×224 image can yield over 2,000 vision tokens, while text inputs typically remain under 200 tokens. Consequently, more than 80% of the sequence consists of vision tokens (Shang et al., 2024; Chen et al., 2024b), leading to quadratic computational cost in self-attention computation and key-value (KV)-cache memory, which in turn hinders the efficiency and scalability of VLMs. To alleviate these costs, model compression techniques such as knowledge distillation (Wang et al., 2022; Cao et al., 2025; Zhang et al., 2024c) and quantization (Wang et al., 2024a; Choi & Kim, 2025; Kim et al., 2024) have been explored, but they mainly reduce parameters rather than sequence length. Thus, vision token redundancy remains a key bottleneck, and addressing it directly is essential for building efficient and scalable VLMs.

Recently, visual-token reduction techniques have been actively studied to address this limitation. Existing token-dropping approaches can be broadly categorized into salience-based and diversity-based methods. Salience-based methods assign an importance score to each visual token using signals such as attention weights, token–text relevance, or learned scoring functions, and retain only the tokens considered most relevant to the downstream task (Arif et al., 2025; Zhang et al., 2024b; Ye et al., 2025). By removing low-scoring tokens before or during language-model inference, these methods shorten the multimodal sequence while attempting to preserve the visual evidence most closely associated with the input instruction.

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This approach is intuitive but relies on dense attention map computation, making it incompatible with efficient kernels such as FlashAttention (Dao et al., 2022) and thus offering limited efficiency gains (Lim. 1). Moreover, selecting tokens solely based on saliency signals can prevent balanced preservation of visual regions, restricting overall image understanding. In particular, (Endo et al., 2024) shows that keeping only high-attention tokens induces position bias, causing over-concentration in certain areas and failing to capture essential regions (Lim. 2). These limitations highlight the need for a token-selection strategy that is both computationally efficient and spatially balanced.

In contrast, diversity-based approaches aim to reduce redundancy and ensure broader contextual coverage by preserving tokens that represent different visual regions. DART (Wen et al., 2025) selects a small set of pivots and retains tokens dissimilar to them, while GreedyPrune (Pei et al., 2025) progressively keeps low-redundancy tokens to achieve a more balanced global representation. These methods preserve contextual information more effectively than saliency-based ones, but still face significant limitations. Pivots are usually chosen from tokens with large norms, without considering similarity among them. As shown in Figure 1-left, high-norm pivots often exhibit strong similarity, leading to redundant selections in neighboring regions (and, interestingly, the opposite case can also occur; see Figure 1-right). This results in over-sampling or exclusion in certain areas, leading to poor coverage (Lim. 3). This phenomenon is supported in Appendix C, which illustrates how overlapping pivots select largely overlapping regions in practice. In addition, these methods are sensitive to manually tuned hyperparameters such as the number of pivots and selection thresholds, whose optimal values vary across models, datasets, and even individual samples (Lim. 4).

To address the four limitations above, we propose **Saliency-Guided Partitioning towards Local Coverage for Importance-Aware Token Dropping (SPLIT)**. *SPLIT* is a simple yet theoretically grounded strategy for efficient token selection that balances saliency and diversity. Our contributions are: (i) **Temporal shift**: we measure token variations across ViT layers to estimate saliency. Unlike attention scores, this is less sensitive to position bias and better captures important regions (solution for Lim. 1&2). (ii) **Local budget allocation**: we partition the image into grids and allocate token budgets by region-level importance, ensuring salient areas receive sufficient tokens and local coverage is preserved (solution for Lim. 3). (iii) **Diversity score**: from the self-similarity matrix, we extract statistics to quantify redundancy. Region-level selection based on this mitigates pivot instability, reduces redundancy, and achieves balanced coverage (solution for Lim. 4). (iv) We show superior performance over prior methods across diverse tasks.



Figure 1. Limitations of DART pivot tokens. The red box marks the first pivot, with the heatmap showing its cosine similarity to other tokens. Subsequent pivots (orange boxes) largely overlap with it, revealing redundancy in the pivot selection process. Additional examples are provided in Appendix C.

2. Related Works

2.1. Vision-Language Models

VLMs (Li et al., 2024b;a; Maaz et al., 2023; Lin et al., 2023; Wang et al., 2024b; Bai et al., 2025) are generally composed of a vision encoder, a projector, and a large language model (LLM) (Chiang et al., 2023; Jiang et al., 2023; Team, 2024). The vision encoder, typically a ViT (Dosovitskiy et al., 2020) or a CLIP-family model (Dosovitskiy et al., 2020; Radford et al., 2021; Zhai et al., 2023; Li et al., 2023a), splits an input image into $P \times P$ patches and converts them into a set of visual tokens $\mathcal{V} = \{\mathbf{v}_1, \dots, \mathbf{v}_M\}$. The projector aligns these tokens to the hidden dimension d_h of the LLM, while text input $\mathcal{T} = \{\mathbf{t}_1, \dots, \mathbf{t}_N\}$ is also embedded into the same space. In general, the number of text tokens N is only a few dozens to hundreds, whereas the number of visual tokens M is much larger, often in the hundreds or thousands in practice. The computational complexity of self-attention grows proportionally with $\mathcal{O}(N+M)^2$. To alleviate this bottleneck, recent work has actively explored *token dropping* methods that selectively remove redundant visual tokens during inference.

2.2. Vision Token Dropping in VLMs

Saliency-based approaches estimate token importance using attention scores or hidden state norms and discard low-importance tokens to reduce computational overhead. HiRED (Arif et al., 2025) preserves key regions with a coarse-to-fine strategy, while FastV (Chen et al., 2024a) selects tokens based on cross-attention scores. PyramidDrop (Xing et al., 2024) exploits the increasing redundancy in deeper layers for dropping, and MustDrop (Liu et al., 2024a) and BTP (Li et al., 2025) extend dropping across multiple stages to balance efficiency and performance. SparseVLM (Zhang et al., 2024b) applies text-guided sparsification to remove irrelevant tokens. Although these methods reflect semantic saliency, they still suffer from limited computational efficiency (Lim. 1) and position bias (Lim. 2).

In contrast, diversity-based approaches focus on reducing redundancy and ensuring that preserved tokens represent diverse visual regions. Beyond Attentive Tokens (Long et al., 2023) merges non-attentive tokens by jointly considering attention and similarity, while DART (Wen et al., 2025) employs duplication-aware pivot selection to remove a large fraction of tokens without sacrificing performance. DivPrune (Ranjbar Alvar et al., 2025) formulates token selection as a max-min diversity problem to guarantee coverage, and GreedyPrune (Pei et al., 2025) combines a non-maximum suppression (NMS)-like strategy to suppress redundancy. SAINT (Jeddi et al., 2025) leverages a similarity graph with adaptive thresholds to balance speed and accuracy. However, these methods often select redundant pivots, leading to over-sampling in some regions (Lim. 3) and under-representation in others, and are sensitive to hyperparameter settings (Lim. 4). In this work, we combine saliency- and diversity-based approaches to complement their weaknesses and preserve optimal coverage. Please refer to Appendix A.3 for additional details on related works.

3. Preliminary

Diversity Token Dropping. A common dropping strategy is to suppress duplication among vision tokens by enforcing diversity constraints. Let $X = \{x_j\}_{j=1}^n$ be the set of visual tokens. Given a set of selected pivots $P \subseteq X$, each pivot $p_i \in P$ defines its non-duplicate set:

$$R_i = \{x_j \in X \mid \text{dup}(p_i, x_j) \leq \tau\}, \quad (1)$$

where $\text{dup}(\cdot, \cdot)$ is typically measured by cosine similarity. The final retained tokens are obtained by combining pivots with their non-duplicate representatives:

$$R = \{p_i \mid p_i \in P\} \cup \{x_j \mid \exists p_i \in P, x_j \in R_i\}, \quad (2)$$

This suppresses near-duplicates, leaving diverse tokens as representatives of X . From a theoretical perspective, recent work (Wen et al., 2025) shows that removing τ -duplicate tokens does not severely harm coverage or model outputs. Specifically, under mild assumptions such as Lipschitz continuity of the transformer with respect to Hausdorff distance (d_H), the deviation between the original and pruned token sets is provably bounded by a constant depending on the embedding norm C and the Lipschitz constant κ as follows:

$$\begin{aligned} d_H(X, R) &\leq \sqrt{2(1-\tau)} C, \\ \|f(X) - f(R)\| &\leq \kappa \sqrt{2(1-\tau)} C \end{aligned} \quad (3)$$

This result indicates that duplicate dropping introduces only a controllable approximation error, thereby justifying the robustness of diversity-based token dropping.

Problem. Diversity-based approaches alleviate saliency bias but remain spatially agnostic. Formally, when partitioning the input into K disjoint regions $X_k = \{x \in X : g(x) = k\}$ with $|X_k| = n_k$ and $\sum_{k=1}^K n_k = n$, duplicated tokens may still concentrate within the same X_k , leaving other regions underrepresented. After diversity-based dropping, the retained set is $R \subseteq X$, with $R_k = R \cap X_k$ and $|R_k| = r_k$. Although such methods reduce redundancy globally, they do not regulate regional distribution. This leads to *region-level imbalance*, where some regions suffer $r_k \ll n_k \cdot \frac{|R|}{n}$ (under-representation) or $r_k = 0$ (empty region). We refer to this phenomenon as *region-level imbalance*, which underscores that global diversity alone does not guarantee local coverage. In particular, as highlighted in Lim. 3, redundant pivots are often selected from neighboring regions, making such an imbalance frequent in practice. Coverage for region k can be quantified via the Hausdorff distance as follows:

$$d_H(X_k, R_k) = \sup_{x \in X_k} \inf_{r \in R_k} \|x - r\|. \quad (4)$$

If $r_k = 0$, then $d_H(X_k, R_k) = \infty$, meaning that the region is entirely uncovered. More generally, when $r_k < N_k(\varepsilon)$, where $N_k(\varepsilon)$ denotes the ε -covering number of X_k , we obtain $d_H(X_k, R_k) > \varepsilon$. This formalizes that region-level imbalance inevitably induces local coverage loss, leaving some regions poorly approximated regardless of the global diversity achieved.

4. Proposed SPLIT

As illustrated in Figure 2, *SPLIT* is a vision token reduction framework that effectively eliminates redundancy while retaining saliency information. It consists of several key components: a temporal shift-based importance estimation and adaptive budget allocation (Section 4.1), which adaptively determines region-wise token counts; a diversity-guided token selection strategy (Section 4.2), which favors representative and informative tokens; a theoretical analysis (Section 4.3), which formally justifies the approach. The overall workflow is summarized in Algorithm 1.

4.1. Temporal Shift Adaptive Allocation

As discussed in Problem. paragraph, suppressing redundancy only at the global level makes it difficult to control imbalance across regions, and tokens in certain areas may disappear entirely, causing severe loss of semantic coverage. To avoid this, we assign importance-based budgets to each region, guaranteeing a minimum level of local coverage. This goes beyond global redundancy suppression by imposing a constraint that ensures every region maintains acceptable approximation accuracy. However, as noted in Lim. 1&2, using attention scores directly as importance

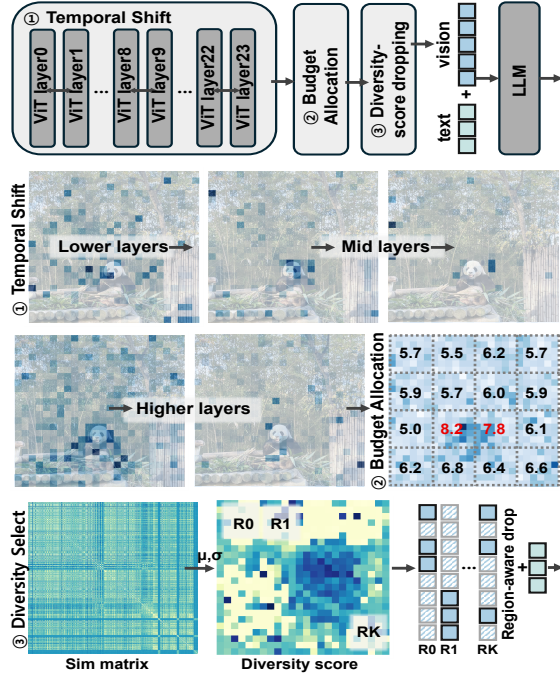


Figure 2. Illustration of the algorithm during three stages: temporal shift quantification, budget allocation, and diversity-based selection.

signals is inefficient, since it requires computing dense attention maps. To address this, we define *temporal shift*—the variation of tokens across layers—as a new importance metric. Specifically, let the hidden state of a token x at layer $\ell \in \{0, \dots, L\}$ be $h_\ell(x) \in \mathbb{R}^d$, and define its relative change rate as follows:

$$\Delta_\ell(x) = \frac{\|h_\ell(x) - h_{\ell-1}(x)\|_2}{\|h_\ell(x)\|_2}, \quad (5)$$

This represents not a simple norm difference but a normalized variation relative to the current representation magnitude, which prevents tokens with large embedding norms from being overemphasized, as shown in Figure 3. In other words, $\Delta_\ell(x)$ quantifies how differently the model reinterprets a token compared to the previous layer, providing an efficient and interpretable importance signal without relying on attention-based saliency. The overall importance of a token is then defined as the average variation across a set of layers $\mathcal{L} \subseteq \{1, \dots, L\}$, and the region-level importance $I(X_k)$ is computed as the average token importance within each region:

$$I(X_k) = \frac{1}{|X_k|} \sum_{x \in X_k} I(x), \quad I(x) = \frac{1}{|\mathcal{L}|} \sum_{\ell \in \mathcal{L}} \Delta_\ell(x), \quad (6)$$

Intuitively, a large $I(x)$ indicates that the model has repeatedly updated and attended to the token, suggesting high

Algorithm 1 SPLIT framework for token reduction

Input: Token set $X = \{x_j\}_{j=1}^n$, partition $X = \bigcup_{k=1}^K X_k$, total budget B , layer set $\mathcal{L}$

Output: Reduced token set $\hat{R} \subseteq X$

- 1: // 1) Compute token importance and 2) budget allocation (See Sec. 4.1)
- 2: Compute hidden states $h_\ell(x)$ for all $x \in X, \ell \in \mathcal{L}$
- 3: Compute token importance and region importance
- 4: $I(x) \leftarrow \frac{1}{|\mathcal{L}|} \sum_{\ell \in \mathcal{L}} \frac{\|h_\ell(x) - h_{\ell-1}(x)\|_2}{\|h_\ell(x)\|_2}$
- 5: $I(X_k) \leftarrow \frac{1}{|X_k|} \sum_{x \in X_k} I(x)$
- 6: Region budget $\tilde{B}_k \leftarrow \frac{B}{K} + B \cdot \frac{I(X_k)}{\sum_{j=1}^K I(X_j)}$
- 7: Round $\tilde{B}_k$ to nearest integer
- 8: // 3) Diversity-based selection (See Sec. 4.2)
- 9: Compute similarity matrix S_{ij} for all $x_i, x_j \in X$
- 10: Compute $\mu_i \leftarrow \text{mean}(S_{i,:})$, $\sigma_i \leftarrow \text{std}(S_{i,:})$
- 11: Diversity score $D(i) \leftarrow \lambda \cdot \sigma_i - \mu_i$
- 12: **for** each region X_k **do**
- 13: choose $\tilde{B}_k$ tokens with largest $D(\cdot)$ in region k
- 14: $\hat{R}_k \leftarrow \arg \max_{R \subseteq X_k, |R|=\tilde{B}_k} \sum_{x \in R} D(x)$
- 15: **end for**
- 16: Final reduced set $\hat{R} \leftarrow \bigcup_{k=1}^K \hat{R}_k$
- 17: **return** $\hat{R}$

informational saliency, whereas a small $I(x)$ implies that the token changes little across layers, likely corresponding to low-information or redundant patterns.

Based on this, the total budget B is distributed across regions. A purely uniform allocation cannot correct inter-region imbalance, while a purely importance-based allocation may overly concentrate on a few regions. Therefore, we adopt a hybrid strategy that combines the two as follows:

$$\tilde{B}_k = \frac{B}{K} + B \cdot \frac{I(X_k)}{\sum_{j=1}^K I(X_j)}. \quad (7)$$

The allocation $\tilde{B}_k$ is then rounded to the nearest integer. As a result, our method achieves two properties: (i) temporal shift-based signals avoid reliance on attention scores, improving efficiency (**Lim. 1**) and reducing sensitivity to position bias (**Lim. 2**); (ii) region-wise budget allocation ensures most regions retain sufficient tokens, mitigating imbalance (**Lim. 3**) and improving local coverage.

4.2. Diversity Score-based Selection

While region-wise budget allocation alleviates imbalance, selecting only the most important tokens within each region can still cause redundancy. Existing greedy methods also aim for diversity but prune sequentially from the global set, making them incompatible with region-specific budgets $\tilde{B}_k$. To overcome this, we introduce *diversity score-based*

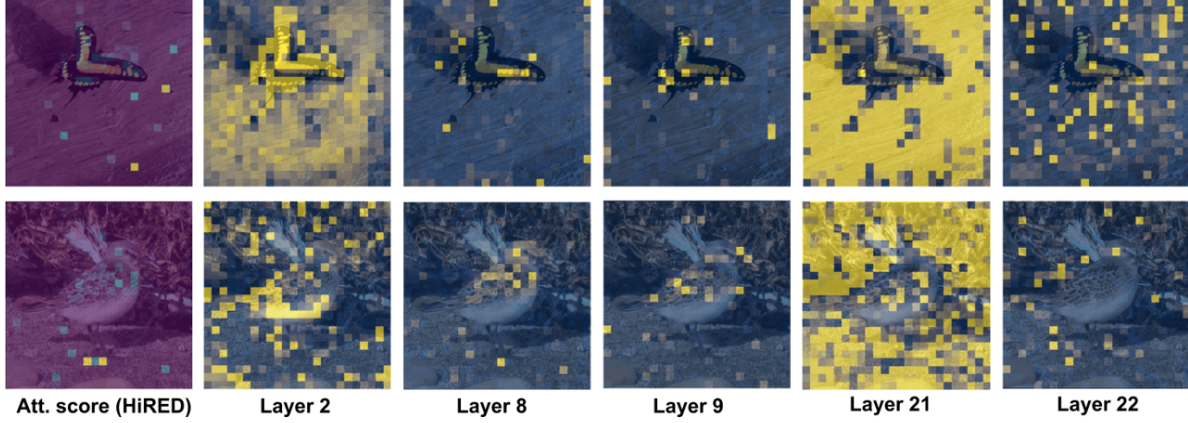


Figure 3. Comparison of attention-score map and temporal shift. Temporal shift highlights salient regions in early–mid layers and broader scene context in later layers, capturing layer-wise representation updates.

selection, which suppresses redundancy within each $\tilde{B}_k$ while ensuring global diversity. Specifically, we normalize each token embedding $x_i \in \mathbb{R}^d$ and define the self-similarity matrix as:

$$S_{ij} = \left\langle \frac{x_i}{\|x_i\|_2}, \frac{x_j}{\|x_j\|_2} \right\rangle, \quad i, j = \{1, \dots, N\}, \quad (8)$$

At this point, the mean similarity and standard deviation of token i are defined as follows:

$$\mu_i = \frac{1}{N} \sum_{j=1}^N S_{ij}, \quad \sigma_i = \sqrt{\frac{1}{N} \sum_{j=1}^N (S_{ij} - \mu_i)^2}, \quad (9)$$

where μ_i represents how much token i is redundantly related to other tokens at the global level, with larger values indicating higher redundancy. In contrast, σ_i denotes the variance of similarities: it is small when the token is uniformly similar to all others, and large when it is strongly associated with only a specific subset. Thus, a larger σ_i implies that the token deviates from homogeneous redundancy and occupies a more distinctive representation. We combine these two indicators to define a *diversity score*, which is then used to select the optimal token set under regional budget constraints as follows:

$$\hat{R}_k = \arg \max_{\tilde{B}_k} \sum_{x \in X_k} D(x), \quad D(i) = \lambda \cdot \sigma_i - \mu_i, \quad (10)$$

Here, $\lambda > 0$ is a weighting hyperparameter (set to 0.5 in *SPLIT*, without tuning across different settings), and $D(i)$ is designed to favor tokens with lower average redundancy and higher distinctiveness. As a result, the proposed regional selection is compatible with $\tilde{B}_k$ while effectively ensuring representational diversity within each region in practice. Moreover, our method does not rely on sensitive thresholds such as the number of pivots or distance cutoffs, thereby effectively mitigating **Lim. 4**.

4.3. Theoretical Analysis

Region Budgets. Let $m = |R|$ denote the total number of tokens to be retained. We assign budgets $B_k \in \mathbb{N}$ with $\sum_{k=1}^K B_k = m$ and retain $R_k \subseteq X_k$ such that $|R_k| = r_k \geq B_k$. This simple constraint prevents empty or severely under-represented regions.

Lemma 1 (Minimum Budget). If $B_k \geq 1$ for all k , then $r_k \geq 1$, hence $d_H(X_k, R_k) < \infty$ for every region. Even a minimal budget eliminates catastrophic coverage failure.

Theorem 1 (Effect of regional budgets). Without regional budgets, some regions may vanish ($r_k = 0$), leading to $d_H(X, R)$ to increase significantly. By Lemma 1, imposing budgets $B_k \geq 1$ ensures $d_H(X, R)$ remains finite.

Lemma 2 (Coverage by covering numbers). If $B_k \geq N_k(\varepsilon)$ for all k , then $d_H(X_k, R_k) \leq \varepsilon$. Consequently $d_H(X, R) = \max_k d_H(X_k, R_k) \leq \varepsilon$.

Definition 1 (Adequacy ratio). Fix $\varepsilon > 0$ and let $N_k(\varepsilon)$ be the ε -covering number of region X_k . define the per-region ratio $\gamma_k(B, \varepsilon) = \frac{B_k}{N_k(\varepsilon)}$, $\gamma_{\min}(B, \varepsilon) = \min_k \gamma_k(B, \varepsilon)$.

Theorem 2 (Adaptive budgets dominate uniform). Let $\gamma_{\min}(B, \varepsilon)$ denote the adequacy ratio (Definition 1). For the uniform allocation $B_k = m/K$ and the adaptive allocation $B_k = \frac{m N_k(\varepsilon)}{\sum_j N_j(\varepsilon)}$,

$$\gamma_{\min}^u(\varepsilon) = \frac{m}{K \max_k N_k(\varepsilon)}, \quad \gamma_{\min}^a(\varepsilon) = \frac{m}{\sum_j N_j(\varepsilon)}. \quad (11)$$

Since $\sum_j N_j(\varepsilon) \leq K \max_k N_k(\varepsilon)$, we have $\gamma_{\min}^a(\varepsilon) \geq \gamma_{\min}^u(\varepsilon)$. Hence, the adaptive allocation achieves the largest adequacy ratio and thus provides a tighter bound on $d_H(X, R)$. Moreover, as noted in Lemma 2, if $B_k \geq N_k(\varepsilon)$ for all k , then each region is ε -covered (i.e., $d_H(X_k, R_k) \leq \varepsilon$), which in turn implies $d_H(X, R) \leq \varepsilon$.

Table 1. Performance comparison conducted with LLaVA-1.5-7B on 10 image understanding tasks. The best-performing results are shown in **bold**, and the second-best results are underlined.

Method	GQA	MMB	MMB-CN	MME	POPE	SQA	VQA ^{txt}	VQA ^{v2}	VizWiz	OCRBench	Avg.
LLaVA-1.5-7B Upper Bound 576 Tokens (100%)											
Vanilla	61.22	63.14	54.81	1477.65	85.41	68.12	48.67	77.41	53.81	205	100%
LLaVA-1.5-7B Retain 192 Tokens ($\downarrow$ 66.7%)											
MustDrop (24.11)	57.1	61.8	53.9	1435.8	82.5	67.9	46.5	73.5	52.7	190.0	96.7%
FastV (ECCV24)	58.2	62.1	54.7	1432.6	81.6	66.6	47.2	71.8	53.0	182.0	96.1%
PDrop (CVPR25)	59.8	60.9	52.0	1455.6	83.2	67.5	46.0	75.9	53.8	185.0	97.5%
HiRED (AAAI25)	58.8	62.4	<u>54.7</u>	1423.8	82.5	66.3	47.2	72.4	52.3	180.0	95.7%
SparseVLM (ICML25)	57.8	62.1	54.8	1443.2	81.5	66.7	47.0	73.3	52.8	182.0	96.6%
DivPrune (CVPR25)	60.5	61.2	52.8	1437.4	85.3	67.2	45.4	<u>76.1</u>	<u>55.1</u>	187.0	96.9%
DART (EMNLP25)	59.2	62.7	53.4	1477.8	84.2	<u>67.9</u>	46.7	76.2	54.6	191.0	99.0%
GreedyPrune (25.01)	58.9	61.2	52.5	1432.1	84.0	<u>67.1</u>	44.6	75.4	54.9	185.0	96.4%
SPLIT (Ours)	59.5	<u>62.6</u>	54.5	1479.5	85.6	68.1	<u>46.9</u>	<u>76.1</u>	55.3	192.0	99.3%
LLaVA-1.5-7B Retain 128 Tokens ($\downarrow$ 77.8%)											
MustDrop (24.11)	56.7	59.0	50.9	1413.8	81.8	65.8	42.6	73.7	53.5	175.0	94.4%
FastV (ECCV24)	55.3	58.3	50.8	1400.5	79.5	63.2	40.2	72.3	52.3	175.0	93.3%
PDrop (CVPR25)	57.1	60.3	50.4	1417.9	81.0	66.7	44.3	73.8	54.4	180.0	95.0%
HiRED (AAAI25)	56.5	59.3	51.5	1401.5	80.2	64.3	40.0	72.8	52.4	179.0	93.7%
SparseVLM (ICML25)	56.3	59.0	51.1	1402.4	80.1	64.4	41.1	71.9	52.5	172.0	93.4%
DivPrune (CVPR25)	60.0	60.2	50.5	1411.9	84.8	67.4	44.0	<u>75.0</u>	<u>55.5</u>	181.0	95.2%
DART (EMNLP25)	57.9	62.2	52.1	1449.2	83.3	68.0	45.2	<u>75.0</u>	54.6	186.0	97.2%
GreedyPrune (25.01)	57.5	60.5	49.7	1411.1	83.8	66.9	44.1	74.5	55.3	180.0	94.9%
SPLIT (Ours)	<u>58.8</u>	62.5	<u>51.9</u>	1455.2	85.5	68.4	45.8	75.5	55.6	188.0	97.8%
LLaVA-1.5-7B Retain 64 Tokens ($\downarrow$ 88.9%)											
MustDrop (24.11)	54.8	54.4	48.4	1329.7	81.0	66.2	40.0	71.4	53.7	158.0	89.2%
FastV (ECCV24)	53.1	50.1	48.6	1292.8	78.5	65.5	39.5	70.3	52.6	150.0	86.6%
PDrop (CVPR25)	54.6	56.0	50.1	1343.2	81.8	66.4	40.4	71.8	54.1	163.0	90.3%
HiRED (AAAI25)	54.0	50.3	48.5	1297.5	78.0	66.3	39.2	70.4	52.1	148.0	86.8%
SparseVLM (ICML25)	53.8	50.5	49.0	1289.3	78.3	65.4	39.6	70.6	52.6	152.0	86.6%
DivPrune (CVPR25)	58.4	58.6	47.5	1362.1	83.7	<u>67.3</u>	40.8	<u>72.4</u>	<u>55.4</u>	166.0	91.7%
DART (EMNLP25)	55.1	59.7	49.2	1357.6	82.6	67.9	40.7	72.1	54.9	167.0	91.4%
GreedyPrune (25.01)	55.0	<u>55.2</u>	45.3	<u>1365.1</u>	82.6	64.6	40.3	71.6	53.5	165.0	91.0%
SPLIT (Ours)	59.0	60.1	<u>49.8</u>	1375.5	84.5	68.6	41.5	73.2	55.9	169.0	92.8%

5. Experiments and Analysis

All experiments were conducted on an NVIDIA H100 GPU. The implementation was developed in Python 3.12 with PyTorch 2.7.0 and CUDA 12.2. For fair comparison, all baseline results were reproduced under the same experimental setup. All experiments were run with a batch size of 1, and none of the models were trained. Further implementation details are provided in Appendix A.

5.1. Main Results

Image understanding results. Table 1 reports results on the LLaVA-1.5-7B model across 10 representative image understanding benchmarks with different token dropping methods. All scores follow the official metrics of each benchmark, and the last column shows the average performance relative to the vanilla model without token dropping. With 192 tokens, *SPLIT* achieves over 99% of the original performance, outperforming the second-best DART by 0.3%. Even under the more extreme 64-token setting, where prior methods suffer severe degradation, *SPLIT* maintains about 92.8% accuracy, surpassing DivPrune by 1.1%. These results demonstrate that by guaranteeing minimal coverage under the same token budget, *SPLIT* preserves information more robustly as the token count decreases, providing strong resilience in highly compressed scenarios.

As shown in Tables 2 and 3, *SPLIT* retains only 11.1% of tokens yet achieves 92.3% on Qwen2-VL and 96.0% on LLaVA-Next. This highlights its ability to mitigate position bias and coverage instability, preserving essential information under tight token budgets with consistent performance across models and tasks. These results further demonstrate the robustness and generalizability of our token-selection strategy. Additional results on more models are provided in Appendix B.

Efficiency Analysis Table 4 presents a quantitative comparison of inference efficiency across different methods. While all approaches achieve some acceleration over the vanilla model, *SPLIT* provides the strongest gains, improving total inference time by $1.80\times$ and the prefilling stage by $2.76\times$. Moreover, it achieves these speedups while keeping FLOPs and KV-cache usage at minimal levels. These improvements arise from *SPLIT*’s ability to prune redundant tokens before entering the LLM and its avoidance of greedy search procedures that repeatedly compute cosine similarities for pivots. Furthermore, as shown in Figure 4, *SPLIT* consistently maintains superior performance regardless of the pruning layer, underscoring its robustness. Overall, these results confirm *SPLIT* as a robust approach that achieves a superior efficiency–accuracy trade-off by leveraging the complementary strengths of saliency and diversity.

Table 2. Comparative results on Qwen2-VL-7B.

Method	GQA	MMB	MMB-CN	MME	POPE	SQA	VQA ^{txt}	Avg.
Qwen2-VL-7B Upper Bound, All Tokens (100%)								
Vanilla	61.5	80.3	79.0	1679.9	88.9	83.5	82.7	100.0%
Qwen2-VL-7B Token Reduction ($\downarrow$ 66.7%)								
DivPrune (CVPR25)	61.0	78.7	77.2	1655.1	88.6	83.1	79.3	98.5%
DART (EMNLP25)	60.1	80.2	76.6	1635.3	88.4	82.9	<u>78.3</u>	97.5%
GreedyPrune (25.01)	61.2	77.8	77.0	1664.0	89.1	82.5	71.9	98.5%
SPLIT (Ours)	<u>61.0</u>	<u>79.3</u>	77.5	<u>1658.3</u>	<u>88.8</u>	<u>83.0</u>	78.1	98.6%
Qwen2-VL-7B Token Reduction ($\downarrow$ 77.8%)								
DivPrune (CVPR25)	60.5	77.3	76.1	1621.7	88.1	<u>82.1</u>	76.2	96.6%
DART (EMNLP25)	58.3	78.2	75.0	1614.7	87.7	82.3	73.4	96.0%
GreedyPrune (25.01)	<u>60.4</u>	75.7	75.0	1637.2	87.8	81.1	71.7	96.9%
SPLIT (Ours)	60.1	<u>77.6</u>	<u>75.9</u>	<u>1636.0</u>	88.1	<u>82.1</u>	<u>74.9</u>	97.2%
Qwen2-VL-7B Token Reduction ($\downarrow$ 88.9%)								
DivPrune (CVPR25)	59.3	75.2	<u>73.0</u>	<u>1542.1</u>	87.9	81.5	66.7	<u>92.1%</u>
DART (EMNLP25)	55.1	74.5	72.3	1491.4	84.5	82.1	<u>68.5</u>	89.4%
GreedyPrune (25.01)	57.5	73.3	72.3	1509.7	85.1	80.7	65.2	90.2%
SPLIT (Ours)	<u>58.0</u>	75.2	73.7	1545.8	<u>86.4</u>	<u>81.7</u>	68.8	92.3%

Table 3. Comparative results on LLaVA-Next-7B.

Method	GQA	MMB	MMB-CN	MME	POPE	SQA	VQA ^{txt}	Avg.
LLaVA-Next-7B Upper Bound, All Tokens (100%)								
Vanilla	62.2	66.3	57.0	1483.5	87.7	69.3	69.3	100.0%
LLaVA-Next-7B Token Reduction ($\downarrow$ 66.7%)								
DivPrune (CVPR25)	<u>61.5</u>	66.0	55.4	1470.6	<u>87.6</u>	69.9	58.7	98.9%
DART (EMNLP25)	61.2	66.0	<u>56.1</u>	<u>1473.6</u>	87.7	69.2	62.6	<u>99.3%</u>
GreedyPrune (25.01)	60.9	66.1	54.2	1457.3	86.8	68.6	57.3	97.9%
SPLIT (Ours)	61.7	66.1	56.2	1479.3	87.5	69.9	62.6	99.6%
LLaVA-Next-7B Token Reduction ($\downarrow$ 77.8%)								
DivPrune (CVPR25)	61.0	65.9	54.1	1438.1	87.3	69.7	55.3	96.9%
DART (EMNLP25)	61.6	<u>65.6</u>	<u>54.6</u>	<u>1462.9</u>	<u>87.2</u>	<u>68.5</u>	61.1	98.5%
GreedyPrune (25.01)	60.2	63.4	54.0	1439.5	86.4	68.2	54.4	96.6%
SPLIT (Ours)	<u>61.2</u>	65.2	54.9	1465.3	87.0	68.4	<u>60.7</u>	98.6%
LLaVA-Next-7B Token Reduction ($\downarrow$ 88.9%)								
DivPrune (CVPR25)	59.8	64.9	52.6	1384.4	<u>86.0</u>	68.8	50.0	93.5%
DART (EMNLP25)	60.5	<u>65.1</u>	<u>53.4</u>	1416.5	85.6	67.8	57.5	<u>95.6%</u>
GreedyPrune (25.01)	59.5	64.8	52.3	1383.8	85.4	67.8	50.4	93.3%
SPLIT (Ours)	<u>60.3</u>	65.2	53.6	<u>1410.9</u>	86.2	<u>68.6</u>	<u>55.4</u>	96.0%

Table 4. Inference costs in LLaVA-Next-7B under 88.9% token compression: number of tokens, total time, prefilling time, FLOPs, KV cache memory, POPE performance, and speedup.

Methods	Tokens $\downarrow$	Total Time $\downarrow$ (Min:Sec)	Prefilling Time $\downarrow$ (Min:Sec)	FLOPs $\downarrow$	KV Cache $\downarrow$ (MB)	POPE $\uparrow$ (F1-Score)	Speedup $\uparrow$ (Total)	Speedup $\uparrow$ (Prefilling)
Vanilla LLaVA-Next-7B	2880	46:34	28:59	100%	1471	87.7	1.00 $\times$	1.00 $\times$
+ DivPrune (CVPR25)	320	26:34	10:57	19.8%	192	86.0	1.75 $\times$	2.65 $\times$
+ DART (EMNLP25)	320	27:28	11:19	24.8%	277	85.6	1.70 $\times$	2.56 $\times$
+ GreedyPrune (25.01)	320	31:17	12:02	22.3%	232	85.4	1.49 $\times$	2.41 $\times$
+ SPLIT (Ours)	320	25:52	10:31	19.8%	192	86.2	1.80$\times$	2.76$\times$

Video understanding results. Video understanding involves far more severe visual redundancy than image-based tasks, making it a representative multimodal problem where token counts grow exponentially. Processing multiple consecutive frames can easily expand the input to tens of thousands of tokens, rendering efficient token management essential for practical deployment. As shown in Table 5, with a 64×32 token budget, *SPLIT* achieves 63.0 on MLVU, surpassing the second-best DivPrune by +1.6. On average accuracy, *SPLIT* retains 87.7% of the vanilla model performance, outperforming DivPrune and DART by +0.9% and +0.4%, respectively. This trend holds consistently across different token ratios, indicating that even in highly redundant video scenarios, *SPLIT* supports practical scalability to large-scale video applications.

5.2. Ablation Study

Region budget matters. To validate our region-level budget allocation strategy, Figure 5 compares the performance of different allocation schemes. Purely random selection, despite being unguided, maintains reasonable performance by preserving regional diversity through dispersed sampling. In contrast, uniform allocation guarantees a minimum budget for each region, thereby ensuring overall coverage and preventing the complete omission of specific regions, which yields more stable and higher performance than random selection. Finally, importance-based adaptive allocation delivers the most notable improvements: while

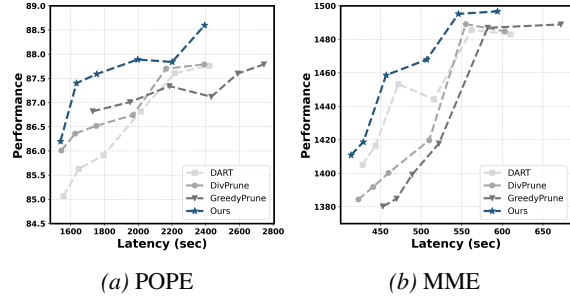


Figure 4. Effect of LLM layer position (0,2,5, 10,15,20) on dropping across different methods in LLaVA-Next-7B, with tokens compressed to 88.9%.

relying on the same random sampling process within regions, adjusting the budget by regional importance consistently outperforms both uniform and random allocation. This result further demonstrates that Temporal Shift provides an effective criterion for region-level budget allocation. Specifically, our adaptive allocation assigns regional budgets according to temporal shift, whereas HiRED relies on attention-based saliency to estimate token importance. Thus, the observed gains can be attributed to the regional budget allocation criterion, rather than to differences in the intra-region selection procedure. Moreover, under constrained settings, the proposed temporal-shift-driven allocation even surpasses saliency-based (*i.e.*, attention-based) methods such as HiRED.

Table 5. Evaluation results on LLaVA-Video-7B across multiple benchmarks.

Method	MLVU	MVBench	LongVideoBench			Video-MME				Avg.
Metric	Avg.	Test	Val	Perception	Relation	W/O sub	Short	Medium	Long	
LLaVA-Video-7B	67.7	58.1	59.0	65.1	53.5	63.6	76.6	61.3	53.1	100.0%
<i>Upper Bound, All 64×169 Tokens (100%)</i>										
<i>Retain 64×64 Tokens ($\downarrow 62.1\%$)</i>										
FastV (ECCV24)	63.8	54.8	56.1	60.8	52.2	61.0	73.1	58.3	51.0	95.2%
DivPrune (CVPR25)	64.1	55.0	58.7	64.2	53.4	61.0	73.3	59.0	50.8	96.7%
DART (EMNLP25)	64.0	55.0	58.6	64.2	53.5	61.5	73.0	58.2	51.0	96.6%
GreedyPrune (25.01)	64.0	54.5	58.4	64.2	52.7	61.3	72.5	57.9	50.8	95.9%
SPLIT (Ours)	64.4	55.1	58.7	64.2	53.4	61.2	73.3	59.0	51.1	96.9%
<i>Retain 64×32 Tokens ($\downarrow 81.1\%$)</i>										
FastV (ECCV24)	58.6	52.1	52.4	56.9	48.4	56.2	63.6	55.3	48.4	88.1%
DivPrune (CVPR25)	61.9	52.5	56.8	62.4	51.4	59.0	72.5	56.9	49.2	93.6%
DART (EMNLP25)	61.5	52.3	57.3	62.8	51.8	59.4	71.7	55.7	49.2	93.5%
GreedyPrune (25.01)	61.0	51.6	56.3	61.8	51.0	60.0	71.9	55.5	48.9	92.8%
SPLIT (Ours)	62.2	52.9	57.3	62.3	51.8	60.0	72.3	56.0	49.4	93.9%
<i>Retain 64×16 Tokens ($\downarrow 90.5\%$)</i>										
FastV (ECCV24)	52.6	46.9	46.6	48.8	44.4	49.8	55.2	50.3	45.5	78.9%
DivPrune (CVPR25)	57.5	47.9	51.5	56.8	46.2	55.5	68.5	53.5	47.4	86.9%
DART (EMNLP25)	58.6	51.5	51.4	56.9	46.0	54.7	67.9	52.5	47.6	87.3%
GreedyPrune (25.01)	58.0	46.8	50.9	55.8	46.0	55.0	68.0	52.8	47.2	86.1%
SPLIT (Ours)	59.1	49.7	51.8	57.1	46.1	56.3	68.3	53.5	47.6	87.7%

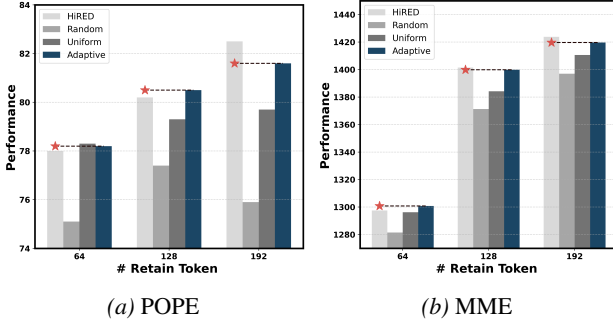


Figure 5. Region-level budget allocation strategies with random sampling using LLaVA-1.5-7B.

Impact of region size. We further analyze the impact of region size on accuracy. Since region-level allocation involves only lightweight computation, latency differences across region sizes are negligible. In contrast, region size has a substantial impact on accuracy, as illustrated in Figure 6a. With fewer regions, each region already contains many tokens, limiting the benefit of allocation. With more regions, fine-grained control improves coverage under larger budgets; for example, 6×6 regions can be beneficial when enough tokens are retained. However, when the budget is small (e.g., 64 tokens retained), excessive partitioning makes the per-region budget too sparse, leading to nearly uniform allocation across regions and consequently reducing accuracy. Considering this trade-off between spatial granularity and per-region token sufficiency, we set the number of regions to 16, which provided the most stable performance in our experiments. This principle can also be extended to models with larger or dynamic visual resolutions (e.g., Qwen-VL family models) by adaptively scaling the region size to preserve comparable token coverage per region.

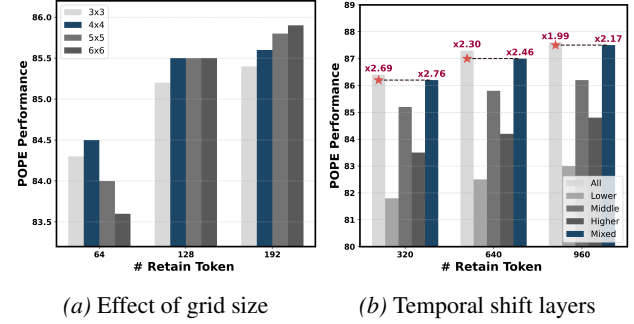


Figure 6. Ablation studies. (a) Effect of different grid-size settings in LLaVA-1.5-7B. (b) Layer selection for applying temporal shift.

Comparison across layer selections. Figure 6b analyzes the effect of choosing different ViT layers when computing temporal shift. Applying temporal shift across all layers yields the best performance, as it comprehensively captures representational changes throughout the network. In contrast, when restricted to lower, middle, or upper layers, the middle layers perform best, likely because they balance low-level visual patterns and high-level semantic information. Moreover, combining three groups consistently outperforms using a single group, further indicating that representations at different levels act in a complementary manner and together provide richer signals than any single level alone. This suggests that temporal shift benefits from multi-level feature changes rather than relying on a specific layer range. The numbers above each bar in Figure 6b denote prefilling speedups, showing that selecting only a subset of layers can improve efficiency compared to using all layers. Therefore, using all layers provides the strongest accuracy, while layer subset selection offers a practical accuracy-efficiency trade-off when prefill overhead is critical. A more detailed discussion is provided in Appendix B.2.

Table 6. LLaVA-1.5-7B results with 64 retained tokens (88.9%). Methods marked with * enforce region-wise token allocation.

Method	GQA	MMB	MMB-CN	MME	POPE	SQA	VQA ^{txt}	VQA ^{v2}	VizWiz	OCRBench	Avg.	M(B)	G(B)
Vispruner	55.1	58.5	49.3	1362.5	83.7	67.1	40.9	72.3	57.0	160	91.4	2.75	0.384
Vispruner*	55.5	58.9	49.7	1365.5	83.9	67.3	41.1	72.8	57.3	162	91.7	2.50	0.358
Visionzip	54.7	58.7	49.7	1353.5	79.4	67.3	40.3	71.3	56.2	155	90.5	2.60	0.377
Visionzip*	55.0	59.4	50.1	1360.4	80.9	67.6	40.6	71.9	56.7	160	91.0	2.50	0.346
CDPruner	58.1	59.7	49.4	1373.1	84.1	67.8	41.7	72.5	55.1	166	92.4	2.45	0.325
CDPruner*	58.9	60.3	50.7	1375.6	84.9	68.1	42.1	73.1	56.3	170	93.0	2.20	0.269

Effects of Region-Level Budgets. To verify the general effect of region-wise budget allocation, we additionally apply the same budget constraint to representative vision token pruning methods, including VisPruner (Zhang et al., 2024a), VisionZip (Yang et al., 2025), and CDPruner (Zhang et al., 2025). Although these methods are not part of the main comparison, they allow us to test whether local coverage constraints benefit different token selection criteria. The constrained variants are marked with * in Table 6. Across all methods, region-wise allocation consistently improves performance. We further quantify regional imbalance using two metrics. Let $B = \{B_1, \dots, B_K\}$ denote the retained token counts across K regions, and let $\bar{B}$ be their mean. We use the Max/Mean ratio, $M(B) = \max_k B_k / \bar{B}$, to measure concentration in the dominant region, and the Gini coefficient, $G(B) = \frac{\sum_{i=1}^K \sum_{j=1}^K |B_i - B_j|}{2K \sum_{k=1}^K B_k}$, to measure over-all allocation inequality. As shown in Table 6, region-wise budgets reduce both metrics across all methods, indicating more balanced token distributions. This demonstrates that local coverage constraints mitigate regional imbalance and improve pruning robustness.

Ablation on Temporal Shift and Diversity Score. To disentangle the contributions of Temporal Shift and the Diversity Score, we conduct an ablation under the same SPLIT pipeline. SPLIT performs token reduction in two stages: region-level budget allocation and token selection. To isolate the role of each signal, we replace the guiding signal of each stage with either Temporal Shift or the Diversity Score. Table 7 reports the results on Qwen2-VL-7B under 88.9% token reduction. Using Temporal Shift for allocation and the Diversity Score for selection achieves the best performance, with 1545.8 MME, 86.4 POPE, 69.4 VQA^{v2}, and 85.6% average accuracy. This indicates that the two signals are complementary: Temporal Shift better estimates region-level importance, while the Diversity Score better preserves non-redundant tokens within the allocated regions.

Applicability to Inner LLM Layers. We also conduct a preliminary LLM-side extension, where pruning is applied to vision tokens after an early LLM layer. This setting examines whether cross-modal interactions inside the LLM can provide additional cues for token reduction beyond the ViT-side signals used in SPLIT. We consider two vari-

Table 7. Controlled ablation of allocation and selection signals on Qwen2-VL-7B under 88.9% token reduction.

Allocation Signal	Selection Signal	MME	POPE	VQA ^{v2}	Avg. (%)
Diversity Score	Temporal Shift	1534.1	86.1	68.5	84.8
Diversity Score	Diversity Score	1537.2	85.9	68.3	85.0
Temporal Shift	Temporal Shift	1541.3	86.2	68.9	85.2
Temporal Shift	Diversity Score	1545.8	86.4	69.4	85.6

Table 8. Preliminary LLM-side extension on Qwen2-VL-7B under 88.9% token reduction. SPLIT-LLM applies pruning to vision tokens after an early LLM layer.

Method	MME	POPE	VQA ^{v2}	Avg. (%)
SPLIT (current)	1545.8	86.4	69.4	85.6
SPLIT-LLM (A)	1543.1	86.7	69.1	85.6
SPLIT-LLM (B)	1553.4	87.3	70.1	86.0

ants: SPLIT-LLM (A), which uses cross-modal similarity between vision and text tokens for region-level budget allocation, and SPLIT-LLM (B), which keeps the original allocation scheme but incorporates cross-modal importance with the Diversity Score during token selection. As shown in Table 8, SPLIT-LLM (A) matches the current SPLIT, while SPLIT-LLM (B) improves the average accuracy from 85.6% to 86.0%. This suggests that LLM-side cross-modal signals can provide complementary information for refining token selection. However, the latency gain is limited because all vision tokens must pass through at least one LLM layer before pruning. Therefore, while LLM-side pruning is feasible and can slightly improve accuracy, the current ViT-side design remains more practical for acceleration.

6. Conclusion

We present *SPLIT*, a principled framework for efficient token dropping in VLMs that combines temporal-shift saliency estimation, adaptive region-level budget allocation, and diversity-aware token selection. This design addresses the limitations of purely saliency- or diversity-based approaches while providing stronger coverage guarantees. Experiments on image and video benchmarks demonstrate state-of-the-art efficiency-performance trade-offs under constrained token budgets, highlighting the potential of *SPLIT* as a scalable and theoretically grounded approach to token allocation in multimodal learning.

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Impact Statement

This paper presents work whose goal is to advance the field of Machine Learning. There are many potential societal consequences of our work, none which we feel must be specifically highlighted here.

References

- Alayrac, J.-B., Donahue, J., Luc, P., Miech, A., Barr, I., Hasson, Y., Lenc, K., Mensch, A., Millican, K., Reynolds, M., et al. Flamingo: a visual language model for few-shot learning. *Advances in neural information processing systems*, 35:23716–23736, 2022.
- Arif, K. H. I., Yoon, J., Nikolopoulos, D. S., Vandieren-donck, H., John, D., and Ji, B. Hired: Attention-guided token dropping for efficient inference of high-resolution vision-language models. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 39, pp. 1773–1781, 2025.
- Bai, S., Chen, K., Liu, X., Wang, J., Ge, W., Song, S., Dang, K., Wang, P., Wang, S., Tang, J., et al. Qwen2.5-vl technical report. *arXiv preprint arXiv:2502.13923*, 2025.
- Bigham, J. P., Jayant, C., Ji, H., Little, G., Miller, A., Miller, R. C., Miller, R., Tatarowicz, A., White, B., White, S., et al. Vizwiz: nearly real-time answers to visual questions. In *Proceedings of the 23rd annual ACM symposium on User interface software and technology*, pp. 333–342, 2010.
- Cao, J., Zhang, Y., Huang, T., Lu, M., Zhang, Q., An, R., Ma, N., and Zhang, S. Move-kd: Knowledge distillation for vlms with mixture of visual encoders. In *Proceedings of the Computer Vision and Pattern Recognition Conference*, pp. 19846–19856, 2025.
- Chaoyou, F., Peixian, C., Yunhang, S., Yulei, Q., Mengdan, Z., Xu, L., Jinrui, Y., Xiawu, Z., Ke, L., Xing, S., et al. Mme: A comprehensive evaluation benchmark for multimodal large language models. *arXiv preprint arXiv:2306.13394*, 3, 2023.
- Chen, L., Zhao, H., Liu, T., Bai, S., Lin, J., Zhou, C., and Chang, B. An image is worth 1/2 tokens after layer 2: Plug-and-play inference acceleration for large vision-language models. In *European Conference on Computer Vision*, pp. 19–35. Springer, 2024a.
- Chen, L., Zhao, H., Liu, T., Bai, S., Lin, J., Zhou, C., and Chang, B. An image is worth 1/2 tokens after layer 2: Plug-and-play inference acceleration for large vision-language models. In *European Conference on Computer Vision*, pp. 19–35. Springer, 2024b.
- Chiang, W.-L., Li, Z., Lin, Z., Sheng, Y., Wu, Z., Zhang, H., Zheng, L., Zhuang, S., Zhuang, Y., Gonzalez, J. E., et al. Vicuna: An open-source chatbot impressing gpt-4 with 90%* chatgpt quality. See <https://vicuna.lmsys.org> (accessed 14 April 2023), 2(3):6, 2023.
- Choi, D. and Kim, H. Gradq-vit: Robust and efficient gradient quantization for vision transformers. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 39, pp. 16019–16027, 2025.
- Dao, T., Fu, D., Ermon, S., Rudra, A., and Ré, C. Flashattention: Fast and memory-efficient exact attention with io-awareness. *Advances in neural information processing systems*, 35:16344–16359, 2022.
- Dosovitskiy, A., Beyer, L., Kolesnikov, A., Weissenborn, D., Zhai, X., Unterthiner, T., Dehghani, M., Minderer, M., Heigold, G., Gelly, S., et al. An image is worth 16x16 words: Transformers for image recognition at scale. *arXiv preprint arXiv:2010.11929*, 2020.
- Endo, M., Wang, X., and Yeung-Levy, S. Feather the throttle: Revisiting visual token pruning for vision-language model acceleration. *arXiv preprint arXiv:2412.13180*, 2024.
- Fu, C., Dai, Y., Luo, Y., Li, L., Ren, S., Zhang, R., Wang, Z., Zhou, C., Shen, Y., Zhang, M., et al. Video-mme: The first-ever comprehensive evaluation benchmark of multi-modal llms in video analysis. In *Proceedings of the Computer Vision and Pattern Recognition Conference*, pp. 24108–24118, 2025.
- Goyal, Y., Khot, T., Summers-Stay, D., Batra, D., and Parikh, D. Making the v in vqa matter: Elevating the role of image understanding in visual question answering. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pp. 6904–6913, 2017.
- Hudson, D. A. and Manning, C. D. Gqa: A new dataset for real-world visual reasoning and compositional question answering. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pp. 6700–6709, 2019.

- Jeddi, A., Baghbanzadeh, N., Dolatabadi, E., and Taati, B. Similarity-aware token pruning: Your vlm but faster. *arXiv preprint arXiv:2503.11549*, 2025.
- Jiang, A. Q., Sablayrolles, A., Mensch, A., Bamford, C., Chaplot, D. S., de Las Casas, D., Bressand, F., Lengyel, G., Lample, G., Saulnier, L., Lavaud, L. R., Lachaux, M.-A., Stock, P., Scao, T. L., Lavril, T., Wang, T., Lacroix, T., and Sayed, W. E. Mistral 7b. *ArXiv*, abs/2310.06825, 2023. URL <https://api.semanticscholar.org/CorpusID:263830494>.
- Kim, N. J., Lee, J., and Kim, H. Hyq: Hardware-friendly post-training quantization for cnn-transformer hybrid networks. In *Proceedings of the Thirty-Third International Joint Conference on Artificial Intelligence, IJCAI-24. International Joint Conferences on Artificial Intelligence Organization*, volume 8, pp. 4291–4299, 2024.
- Li, B., Zhang, Y., Guo, D., Zhang, R., Li, F., Zhang, H., Zhang, K., Zhang, P., Li, Y., Liu, Z., et al. Llava-onevision: Easy visual task transfer. *arXiv preprint arXiv:2408.03326*, 2024a.
- Li, F., Zhang, R., Zhang, H., Zhang, Y., Li, B., Li, W., Ma, Z., and Li, C. Llava-next-interleave: Tackling multi-image, video, and 3d in large multimodal models. *arXiv preprint arXiv:2407.07895*, 2024b.
- Li, J., Li, D., Savarese, S., and Hoi, S. Blip-2: Bootstrapping language-image pre-training with frozen image encoders and large language models. In *International conference on machine learning*, pp. 19730–19742. PMLR, 2023a.
- Li, K., Wang, Y., He, Y., Li, Y., Wang, Y., Liu, Y., Wang, Z., Xu, J., Chen, G., Luo, P., et al. Mvbench: A comprehensive multi-modal video understanding benchmark. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 22195–22206, 2024c.
- Li, K., Chen, X., Gao, C., Li, Y., and Chen, X. Balanced token pruning: Accelerating vision language models beyond local optimization. *arXiv preprint arXiv:2505.22038*, 2025.
- Li, Y., Du, Y., Zhou, K., Wang, J., Zhao, W. X., and Wen, J.-R. Evaluating object hallucination in large vision-language models. *arXiv preprint arXiv:2305.10355*, 2023b.
- Lin, B., Ye, Y., Zhu, B., Cui, J., Ning, M., Jin, P., and Yuan, L. Video-llava: Learning united visual representation by alignment before projection. *arXiv preprint arXiv:2311.10122*, 2023.
- Liu, H., Li, C., Wu, Q., and Lee, Y. J. Visual instruction tuning. *Advances in neural information processing systems*, 36:34892–34916, 2023.
- Liu, T., Shi, L., Hong, R., Hu, Y., Yin, Q., and Zhang, L. Multi-stage vision token dropping: Towards efficient multimodal large language model. *arXiv preprint arXiv:2411.10803*, 2024a.
- Liu, Y., Duan, H., Zhang, Y., Li, B., Zhang, S., Zhao, W., Yuan, Y., Wang, J., He, C., Liu, Z., et al. Mmbench: Is your multi-modal model an all-around player? In *European conference on computer vision*, pp. 216–233. Springer, 2024b.
- Liu, Y., Li, Z., Huang, M., Yang, B., Yu, W., Li, C., Yin, X.-C., Liu, C.-L., Jin, L., and Bai, X. Ocrbench: on the hidden mystery of ocr in large multimodal models. *Science China Information Sciences*, 67(12):220102, 2024c.
- Long, S., Zhao, Z., Pi, J., Wang, S., and Wang, J. Beyond attentive tokens: Incorporating token importance and diversity for efficient vision transformers. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 10334–10343, 2023.
- Lu, P., Mishra, S., Xia, T., Qiu, L., Chang, K.-W., Zhu, S.-C., Tafjord, O., Clark, P., and Kalyan, A. Learn to explain: Multimodal reasoning via thought chains for science question answering. *Advances in Neural Information Processing Systems*, 35:2507–2521, 2022.
- Maaz, M., Rasheed, H., Khan, S., and Khan, F. S. Video-chatgpt: Towards detailed video understanding via large vision and language models. *arXiv preprint arXiv:2306.05424*, 2023.
- Pei, R., Sun, W., Fu, Z., and Wang, J. Greedyprune: Retenting critical visual token set for large vision language models. *arXiv preprint arXiv:2506.13166*, 2025.
- Radford, A., Kim, J. W., Hallacy, C., Ramesh, A., Goh, G., Agarwal, S., Sastry, G., Askell, A., Mishkin, P., Clark, J., et al. Learning transferable visual models from natural language supervision. In *International conference on machine learning*, pp. 8748–8763. PmLR, 2021.
- Ranjbar Alvar, S., Singh, G., Akbari, M., and Zhang, Y. Divprune: Diversity-based visual token pruning for large multimodal models. *arXiv e-prints*, pp. arXiv–2503, 2025.
- Sapkota, R., Cao, Y., Roumeliotis, K. I., and Karkee, M. Vision-language-action models: Concepts, progress, applications and challenges. *arXiv preprint arXiv:2505.04769*, 2025.
- Shang, Y., Cai, M., Xu, B., Lee, Y. J., and Yan, Y. Llava-prumerge: Adaptive token reduction for efficient large multimodal models. *arXiv preprint arXiv:2403.15388*, 2024.

- Singh, A., Natarajan, V., Shah, M., Jiang, Y., Chen, X., Batra, D., Parikh, D., and Rohrbach, M. Towards vqa models that can read. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pp. 8317–8326, 2019.
- Team, Q. Qwen2 technical report. *arXiv preprint arXiv:2407.10671*, 2024.
- Wang, C., Wang, Z., Xu, X., Tang, Y., Zhou, J., and Lu, J. Q-vlm: Post-training quantization for large vision-language models. *Advances in Neural Information Processing Systems*, 37:114553–114573, 2024a.
- Wang, P., Bai, S., Tan, S., Wang, S., Fan, Z., Bai, J., Chen, K., Liu, X., Wang, J., Ge, W., et al. Qwen2-vl: Enhancing vision-language model’s perception of the world at any resolution. *arXiv preprint arXiv:2409.12191*, 2024b.
- Wang, T., Zhou, W., Zeng, Y., and Zhang, X. Efficientvlm: Fast and accurate vision-language models via knowledge distillation and modal-adaptive pruning. *arXiv preprint arXiv:2210.07795*, 2022.
- Wen, Z., Gao, Y., Wang, S., Zhang, J., Zhang, Q., Li, W., He, C., and Zhang, L. Stop looking for important tokens in multimodal language models: Duplication matters more. *arXiv preprint arXiv:2502.11494*, 2025.
- Wu, H., Li, D., Chen, B., and Li, J. Longvideobench: A benchmark for long-context interleaved video-language understanding. *Advances in Neural Information Processing Systems*, 37:28828–28857, 2024.
- Xing, L., Huang, Q., Dong, X., Lu, J., Zhang, P., Zang, Y., Cao, Y., He, C., Wang, J., Wu, F., et al. Pyramid-drop: Accelerating your large vision-language models via pyramid visual redundancy reduction. *arXiv preprint arXiv:2410.17247*, 2024.
- Yang, S., Chen, Y., Tian, Z., Wang, C., Li, J., Yu, B., and Jia, J. Visionzip: Longer is better but not necessary in vision language models. In *Proceedings of the Computer Vision and Pattern Recognition Conference*, pp. 19792–19802, 2025.
- Ye, X., Gan, Y., Ge, Y., Zhang, X.-P., and Tang, Y. Atp-llava: Adaptive token pruning for large vision language models. In *2025 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 24972–24982. IEEE, 2025.
- Zhai, X., Mustafa, B., Kolesnikov, A., and Beyer, L. Sigmoid loss for language image pre-training. In *Proceedings of the IEEE/CVF international conference on computer vision*, pp. 11975–11986, 2023.
- Zhang, Q., Cheng, A., Lu, M., Zhang, R., Zhuo, Z., Cao, J., Guo, S., She, Q., and Zhang, S. Beyond text-visual attention: Exploiting visual cues for effective token pruning in vlms. *arXiv preprint arXiv:2412.01818*, 2024a.
- Zhang, Q., Liu, M., Li, L., Lu, M., Zhang, Y., Pan, J., She, Q., and Zhang, S. Beyond attention or similarity: Maximizing conditional diversity for token pruning in mllms. *arXiv preprint arXiv:2506.10967*, 2025.
- Zhang, Y., Fan, C.-K., Ma, J., Zheng, W., Huang, T., Cheng, K., Gudovskiy, D., Okuno, T., Nakata, Y., Keutzer, K., et al. Sparsevlm: Visual token sparsification for efficient vision-language model inference. *arXiv preprint arXiv:2410.04417*, 2024b.
- Zhang, Z., Meyer, G. P., Lu, Z., Shrivastava, A., Ravichandran, A., and Wolff, E. M. Vlm-kd: Knowledge distillation from vlm for long-tail visual recognition. *arXiv preprint arXiv:2408.16930*, 2024c.
- Zhou, J., Shu, Y., Zhao, B., Wu, B., Liang, Z., Xiao, S., Qin, M., Yang, X., Xiong, Y., Zhang, B., et al. Mlvu: Benchmarking multi-task long video understanding. In *Proceedings of the Computer Vision and Pattern Recognition Conference*, pp. 13691–13701, 2025a.
- Zhou, W., Tao, M., Zhao, C., Guo, H., Dong, H., Tang, M., and Wang, J. Physvlm: Enabling visual language models to understand robotic physical reachability. In *Proceedings of the Computer Vision and Pattern Recognition Conference*, pp. 6940–6949, 2025b.

Appendix

A. Experimental Setup

A.1. Model Architectures

LLaVA Models. LLaVA-1.5 (Liu et al., 2023) is an open-source multimodal model that combines a CLIP-based vision encoder with the Vicuna language model. Images at 336×336 resolution are mapped into 576 visual tokens. Despite its simple architecture and modest training cost, it delivers strong performance on diverse vision–language tasks and has become a standard benchmark, making it well-suited for evaluating token pruning methods.

LLaVA-Next (Li et al., 2024b) extends the input resolution to 672×672, producing 2,880 visual tokens. The higher resolution improves performance on complex scene understanding and OCR tasks, but at the cost of substantially higher computation. This makes it a key model for assessing how well token compression methods can reduce computation while preserving accuracy.

LLaVA-Video (Lin et al., 2023) extends the LLaVA framework to video understanding by tokenizing every frame. This dramatically increases the input length—often to hundreds of thousands of tokens—enabling long-range temporal reasoning but incurring extreme computational and memory costs. It thus serves as a critical benchmark for evaluating pruning methods under large-scale visual inputs and temporal reasoning scenarios.

Qwen-VL Models. Qwen2-VL (Wang et al., 2024b), built on the Qwen (Team, 2024) language model family, can process up to 1,296 visual tokens. Trained on large-scale multimodal datasets, it achieves strong performance across diverse visual question answering tasks, with support for long contexts and large parameter scales. In our study, we use Qwen2-VL to examine the robustness of pruning methods in large-model settings.

Qwen2.5-VL (Bai et al., 2025), an improved successor to Qwen2-VL, introduces architectural enhancements such as window attention, SwiGLU, and RMSNorm, enabling efficient handling of high-resolution inputs and long sequences. Released in multiple scales (7B, 32B, 72B), it shows particular strength in document understanding tasks. As a state-of-the-art multimodal model, it serves as an important benchmark for assessing the performance–efficiency trade-off in pruning experiments.

A.2. Datasets

Our experiments were conducted across a diverse set of benchmarks designed to evaluate multiple aspects of multimodal intelligence. These datasets span different modalities and task types, enabling us to assess how well the proposed method generalizes to visual understanding, linguistic reasoning, and temporal context processing. For image understanding, we include ten widely used benchmarks: GQA (Hudson & Manning, 2019), MMB and MMB-CN (Liu et al., 2024b), MME (Chaoyou et al., 2023), POPE (Li et al., 2023b), VizWiz (Bigham et al., 2010), SQA (Lu et al., 2022), VQA^{v2} (Goyal et al., 2017), VQA^{txt} (Singh et al., 2019), and OCRBench (Liu et al., 2024c). Together, these datasets cover evaluation dimensions such as object recognition, scene understanding, OCR, and scientific reasoning. For video understanding, we use four benchmarks—MLVU (Zhou et al., 2025a), MVBench (Li et al., 2024c), LongVideoBench (Wu et al., 2024), and Video-MME (Fu et al., 2025)—to measure the model’s ability to perform temporal reasoning and maintain long-context comprehension.

A.3. Comparison Methods

To validate the effectiveness of our approach, we compare it against a set of recently proposed token pruning and optimization methods.

MustDrop. (Liu et al., 2024a) introduces a multi-stage token pruning framework spanning the vision encoding, prefilling, and decoding phases. During vision encoding, adjacent or redundant tokens are merged to form a compact set of core tokens. In the prefilling stage, dual-attention is applied to discard tokens irrelevant to the textual context. Finally, the decoding stage compresses the KV cache to reduce memory usage. This design enables more precise token selection than single-stage approaches, maintaining stable performance even under high compression ratios.

FastV. (Chen et al., 2024a) builds on the observation that many visual tokens in LVLMs contribute little in deeper layers. It dynamically prunes tokens with low attention scores, reducing not only sparse computation but also the actual token

count, thereby lowering overall FLOPs and latency. Experiments show that FastV achieves substantial efficiency gains with minimal performance degradation.

PDrop. (Xing et al., 2024) introduces a progressive token pruning strategy that leverages the hierarchical structure of vision–language models. Empirically, all tokens are needed in shallow layers, while redundancy increases in deeper ones. Accordingly, the model is divided into multiple stages, and a predefined fraction of tokens is dropped at the end of each stage. This preserves information in early layers while maximizing efficiency at greater depths.

HiRed. (Arif et al., 2025) addresses the excessive token issue in high-resolution VLMs. An input image is divided into sub-regions, and CLS-attention from the Vision Transformer allocates a token budget per region, retaining only the most informative tokens. This early token reduction requires no retraining or architectural changes, offering significant efficiency gains while preserving accuracy.

SparseVLM. (Zhang et al., 2024b) improves efficiency by pruning visual tokens that contribute little to vision–language interactions. It exploits the correlation between textual and visual inputs, discarding tokens with low query relevance early in the pipeline. Unlike methods that assess token importance independently, SparseVLM jointly considers vision–language interactions for more precise selection, reducing unnecessary computation and improving memory and latency efficiency. It has been shown to lower GPU memory usage and accelerate inference across multiple multimodal benchmarks.

DivPrune. (Ranjbar Alvar et al., 2025) formulates token pruning as a Max–Min Diversity Problem, selecting tokens that maximize diversity and minimize redundancy. This preserves a more representative subset of the visual input while discarding duplicates. Without requiring retraining or additional tuning, DivPrune achieves state-of-the-art results on a variety of image and video benchmarks.

DART. (Wen et al., 2025) shifts the focus from token importance to token duplication. A subset of tokens is chosen as pivots, and those with high similarity to a pivot are removed, minimizing information loss. The method is simple, fully compatible with FlashAttention, and avoids the attention bias and inefficiency issues of importance-based approaches, consistently outperforming random pruning.

GreedyPrune. (Pei et al., 2025) formulates token pruning as a combinatorial optimization problem, applying a greedy algorithm to select tokens. The key idea is to jointly optimize semantic saliency and visual diversity, mitigating the bias of saliency-based methods while preventing the loss of critical tokens often seen in diversity-based approaches.

Limitations. While prior methods reduce token counts effectively, they still face key limitations: (i) *Limited computational efficiency*: saliency-based approaches rely on dense attention, making them incompatible with efficient kernels such as FlashAttention; (ii) *Position bias*: when pruning is guided solely by attention scores, tokens often cluster in specific regions, restricting global understanding; (iii) *Unstable coverage*: diversity-based approaches insufficiently control overlap among pivots, leading to over-sampling in some regions and neglect in others; (iv) *Hyperparameter sensitivity*: performance depends heavily on choices such as pivot count and thresholds, which vary across models and datasets. Motivated by these observations, we propose *SPLIT*, a new method that jointly ensures both importance and diversity to overcome these limitations.

B. Extended Experimental Results

B.1. Performance across Various Models

In this section, we present extended results on large-scale models beyond those covered in the main paper, aiming to systematically verify the effectiveness of *SPLIT* across diverse architectures and parameter scales. Specifically, we evaluate LLaVA-1.5-13B (Table 9), LLaVA-Next-7B/13B (Table 10, 11), Qwen2-VL-7B (Table 12), and Qwen2.5-VL models at 7B/32B/72B scales (Table 13, 14, 15), observing performance trends under varying token budgets. Across all families and scales, the trends remain consistent, indicating that *SPLIT* is broadly applicable rather than tied to a particular model size. This consistency further demonstrates its practical value, ensuring reliable performance improvements across diverse model choices in real-world applications.

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Table 9. Performance comparison conducted with *LLaVA-1.5-13B* on 10 image understanding tasks. The best score is marked in **bold**, and the second best is marked with underline.

Method	GQA	MMB	MMB-CN	MME	POPE	SQA	VQA (txt)	VQA (v2)	VizWiz	OCRBench	Ratio
LLaVA-1.5-13B Upper Bound 576 Tokens (100%)											
Vanilla	62.6	68.2	61.1	1542.8	85.8	71.6	52.9	78.9	56.6	225	100%
LLaVA-1.5-13B Retain 196 Tokens ($\downarrow 66.7\%$)											
FastV (ECCV24)	59.1	60.4	53.5	1501.0	81.5	68.4	43.5	73.8	52.6	200	95.2 %
PDrop (CVPR25)	60.8	65.8	58.0	1515.5	83.5	70.8	47.2	76.0	54.8	205	97.0 %
HiRED (AAAI25)	59.2	61.0	52.5	1500.8	81.5	67.7	42.5	73.6	51.8	200	95.0%
SparseVLM (ICML25)	59.2	62.9	53.9	1505.2	82.3	68.6	44.5	74.2	53.2	201	95.6%
DivPrune (CVPR25)	61.6	66.9	<u>59.6</u>	1541.4	86.1	71.8	<u>49.6</u>	77.2	<u>56.4</u>	207	98.8%
DART (EMNLP25)	61.0	68.7	60.0	1510.9	83.5	71.9	50.8	77.0	56.7	218	98.0%
GreedyPrune (25.01)	61.3	64.8	58.1	1512.5	85.9	71.0	47.0	76.0	57.0	201	96.9%
SPLIT (Ours)	60.9	<u>67.9</u>	58.8	<u>1532.4</u>	86.1	70.7	<u>47.2</u>	77.2	<u>56.3</u>	<u>210</u>	<u>98.4%</u>
LLaVA-1.5-13B Retain 128 Tokens ($\downarrow 77.8\%$)											
FastV (ECCV24)	58.4	62.1	52.5	1497.5	79.4	64.6	42.2	71.9	52.3	190	94.2%
PDrop (CVPR25)	59.8	65.2	57.3	1507.4	82.5	70.6	46.8	74.6	56.3	199	96.3%
HiRED (AAAI25)	58.3	61.5	51.5	1495.8	78.1	66.8	41.2	70.9	50.6	190	93.9%
SparseVLM (ICML25)	58.3	62.6	53.8	1499.2	80.0	68.2	43.2	73.0	53.8	192	94.7%
DivPrune (CVPR25)	61.1	65.5	<u>58.4</u>	<u>1519.4</u>	85.9	<u>71.7</u>	<u>47.9</u>	<u>75.5</u>	<u>57.0</u>	199	<u>97.2%</u>
DART (EMNLP25)	59.6	67.5	58.9	1510.3	81.8	72.6	48.7	75.3	56.8	205	97.0%
GreedyPrune (25.01)	60.2	65.2	58.0	1500.5	84.2	70.9	46.3	74.2	57.3	200	96.1%
SPLIT (Ours)	59.8	<u>67.3</u>	58.2	1526.3	<u>85.7</u>	71.1	46.7	76.4	54.8	205	97.6%
LLaVA-1.5-13B Retain 64 Tokens ($\downarrow 88.9\%$)											
FastV (ECCV24)	50.0	60.5	50.2	1435.8	74.3	66.9	40.2	70.5	50.4	165	89.5%
PDrop (CVPR25)	56.5	52.8	55.2	1449.4	78.8	71.1	42.5	73.1	55.4	173	91.4%
HiRED (AAAI25)	48.5	60.5	49.4	1431.6	74.9	65.6	40.6	71.2	49.0	160	89.0%
SparseVLM (ICML25)	52.1	60.6	51.2	1440.2	75.7	67.9	40.8	71.7	51.8	165	90.1%
DivPrune (CVPR25)	59.8	63.5	<u>56.6</u>	1456.4	84.7	<u>71.8</u>	43.9	<u>73.9</u>	56.8	181	93.2%
DART (EMNLP25)	57.2	65.0	57.0	1459.0	78.9	73.0	42.1	74.0	<u>56.5</u>	174	92.7%
GreedyPrune (25.01)	56.8	62.3	55.3	1443.1	77.5	71.5	41.2	72.6	<u>56.2</u>	173	91.5%
SPLIT (Ours)	<u>58.3</u>	<u>64.5</u>	56.4	1465.9	78.7	70.9	<u>42.8</u>	73.2	55.1	185	93.3%

Table 10. Performance comparison conducted with *LLaVA-Next-7B* on 10 image understanding tasks.

Method	GQA	MMB	MMB-CN	MME	POPE	SQA	VQA (txt)	VQA (v2)	VizWiz	OCRBench	Avg.
LLaVA-Next-7B Upper Bound 2880 Tokens (100%)											
Vanilla	62.2	66.3	57.0	1483.5	87.7	69.3	64.0	80.1	58.5	492	100%
LLaVA-Next-7B Retain 960 Tokens ($\downarrow 66.7\%$)											
FastV (ECCV24)	59.8	64.8	52.5	1454.0	85.9	67.4	57.0	76.1	55.4	435	95.5 %
PDrop (CVPR25)	60.7	65.6	54.7	1463.7	86.0	68.8	58.9	77.7	57.0	454	97.1 %
HiRED (AAAI25)	60.1	65.0	53.0	1450.8	85.4	67.3	56.6	76.1	55.5	430	95.2%
SparseVLM (ICML25)	59.5	64.5	52.8	1456.9	85.9	67.3	57.1	76.0	55.1	444	96.0%
DivPrune (CVPR25)	<u>61.5</u>	66.0	55.4	1470.6	<u>87.6</u>	69.9	58.7	<u>78.3</u>	58.0	449	97.4%
DART (EMNLP25)	61.2	66.0	<u>56.1</u>	<u>1473.6</u>	87.7	69.2	62.6	78.1	58.2	479	98.9%
GreedyPrune (25.01)	60.9	66.1	54.2	1457.3	86.8	68.6	57.3	77.5	56.1	446	96.4%
SPLIT (Ours)	61.7	66.1	56.2	1479.3	87.5	69.9	62.6	78.5	<u>58.0</u>	<u>475</u>	99.0%
LLaVA-Next-7B Retain 640 Tokens ($\downarrow 77.8\%$)											
FastV (ECCV24)	58.9	63.4	51.2	1425.2	85.3	67.7	54.2	74.2	55.2	425	93.6%
PDrop (CVPR25)	60.5	64.4	53.6	1441.8	86.7	68.5	55.3	76.7	56.2	430	95.0%
HiRED (AAAI25)	59.0	63.1	51.4	1420.9	86.1	67.9	55.0	75.2	55.1	423	93.5%
SparseVLM (ICML25)	58.8	63.2	51.4	1430.8	85.4	67.4	54.9	74.8	55.0	427	94.0%
DivPrune (CVPR25)	61.0	65.9	54.1	1438.1	87.3	69.7	55.3	76.9	57.3	446	95.7%
DART (EMNLP25)	61.6	<u>65.6</u>	<u>54.6</u>	<u>1462.9</u>	<u>87.2</u>	<u>68.5</u>	61.1	77.0	57.8	473	98.0%
GreedyPrune (25.01)	60.3	63.4	54.0	1439.5	86.4	68.2	54.4	75.9	55.9	420	94.3%
SPLIT (Ours)	<u>61.2</u>	65.2	54.9	1465.3	87.0	68.4	<u>60.7</u>	<u>76.9</u>	<u>57.4</u>	<u>470</u>	<u>97.9%</u>
LLaVA-Next-7B Retain 320 Tokens ($\downarrow 88.9\%$)											
FastV (ECCV24)	57.4	52.0	50.6	1381.4	82.1	65.4	50.5	72.3	53.4	357	88.2%
PDrop (CVPR25)	59.4	63.9	52.0	1391.4	84.8	67.2	52.2	74.8	55.7	385	90.7%
HiRED (AAAI25)	57.4	62.1	50.5	1380.5	82.5	65.2	50.4	72.0	53.7	360	88.6%
SparseVLM (ICML25)	57.9	62.0	50.0	1382.1	82.3	65.8	50.5	72.7	53.2	355	88.5%
DivPrune (CVPR25)	59.8	64.9	52.6	1384.4	86.0	68.8	50.0	75.3	56.7	390	90.8%
DART (EMNLP25)	60.5	<u>65.1</u>	53.4	1416.5	85.6	67.8	57.5	76.0	57.4	407	93.1%
GreedyPrune (25.01)	59.5	64.8	52.3	1382.8	85.4	67.8	50.4	73.4	55.6	394	90.7%
SPLIT (Ours)	<u>60.3</u>	65.2	53.6	<u>1410.9</u>	86.2	<u>68.6</u>	<u>55.4</u>	<u>75.8</u>	<u>56.9</u>	420	93.4%

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Table 11. Performance comparison conducted with *LLaVA-Next-13B* on 10 image understanding tasks. The best score is marked in **bold**, and the second best is marked with underline.

Method	GQA	MMB	MMB-CN	MME	POPE	SQA	VQA (txt)	VQA (v2)	VizWiz	OCRBench	Avg.
LLaVA-Next-13B Upper Bound 2880 Tokens (100%)											
Vanilla	64.1	68.4	61.1	1576.2	87.4	72.7	66.4	83.0	63.9	505	100%
LLaVA-Next-13B Retain 960 Tokens ($\downarrow 66.7\%$)											
FastV (ECCV24)	61.5	64.1	58.5	1544.1	85.2	71.7	58.5	80.4	61.6	455	95.9 %
PDrop (CVPR25)	62.8	66.3	57.7	1545.3	86.1	71.2	58.7	81.4	62.5	463	96.5 %
HiRED (AAAI25)	62.2	64.9	59.0	1543.1	85.9	71.5	59.1	81.0	61.9	456	96.1 %
SparseVLM (ICML25)	61.7	64.7	57.9	1545.6	85.8	71.0	58.6	80.6	61.4	460	96.2 %
DivPrune (CVPR25)	63.5	67.0	59.3	1560.4	87.1	72.3	60.2	82.1	<u>63.1</u>	457	97.1 %
DART (EMNLP25)	<u>63.3</u>	68.0	<u>60.1</u>	1560.0	87.0	<u>72.8</u>	65.2	82.1	64.2	<u>491</u>	<u>98.7 %</u>
GreedyPrune (25.01)	62.8	65.7	57.7	1524.1	86.5	72.5	54.8	81.6	61.5	450	95.0 %
SPLIT (Ours)	63.1	68.0	60.9	1563.7	87.1	73.1	<u>64.9</u>	81.1	62.8	495	98.9%
LLaVA-Next-13B Retain 640 Tokens ($\downarrow 77.8\%$)											
FastV (ECCV24)	61.5	63.2	55.9	1524.2	83.5	70.4	57.1	78.1	60.2	420	93.4%
PDrop (CVPR25)	62.2	65.2	56.9	1528.4	85.6	70.9	56.6	78.9	60.7	415	93.7%
HiRED (AAAI25)	61.5	63.9	55.6	1523.4	83.8	70.2	56.1	79.2	61.1	429	93.8%
SparseVLM (ICML25)	61.0	63.3	55.2	1526.0	83.2	70.2	56.9	78.8	60.7	428	93.8%
DivPrune (CVPR25)	62.9	66.6	58.1	1509.2	86.9	72.1	62.5	<u>80.6</u>	61.9	420	93.7%
DART (EMNLP25)	62.4	67.3	59.0	1547.9	86.4	72.0	64.1	81.1	63.9	466	97.1%
GreedyPrune (25.01)	62.4	63.9	55.7	1530.3	86.3	<u>72.4</u>	<u>62.9</u>	79.1	60.5	435	94.7%
SPLIT (Ours)	<u>62.5</u>	<u>66.7</u>	<u>58.9</u>	<u>1545.5</u>	85.1	72.6	62.1	78.3	<u>62.1</u>	470	<u>96.8%</u>
LLaVA-Next-13B Retain 320 Tokens ($\downarrow 88.9\%$)											
FastV (ECCV24)	59.1	61.2	54.2	1475.3	75.1	69.5	49.2	77.5	57.2	356	88.1%
PDrop (CVPR25)	60.5	64.0	55.1	1479.1	81.7	71.3	56.8	78.1	59.4	396	90.7%
HiRED (AAAI25)	59.5	62.1	54.0	1477.3	75.2	70.1	49.6	78.3	57.9	355	88.3%
SparseVLM (ICML25)	58.9	61.9	54.2	1474.1	75.1	69.8	49.2	77.4	57.5	360	88.3%
DivPrune (CVPR25)	61.2	<u>64.8</u>	<u>55.9</u>	1481.1	85.8	<u>71.8</u>	<u>58.2</u>	78.6	59.6	404	91.4%
DART (EMNLP25)	61.0	65.9	57.6	1481.2	<u>84.5</u>	71.7	60.3	79.1	62.2	425	92.5%
GreedyPrune (25.01)	<u>60.3</u>	63.6	54.6	1471.5	81.3	72.1	<u>58.2</u>	77.1	58.0	397	90.4%
SPLIT (Ours)	60.5	63.9	55.8	1483.1	83.5	71.2	56.7	<u>78.7</u>	59.3	<u>420</u>	<u>91.9%</u>

Table 12. Performance comparison conducted with *Qwen2-VL-7B* on 10 image understanding tasks.

Method	GQA	MMB	MMB-CN	MME	POPE	SQA	VQA (txt)	VQA (v2)	VizWiz	OCRBench	Avg.
Qwen2-VL-7B Upper Bound All Tokens (100%)											
Vanilla	61.5	80.3	79.0	1679.9	88.9	83.5	82.7	73.4	68.6	801	100%
Qwen2-VL-7B Token Reduction ($\downarrow 66.7\%$)											
FastV (ECCV24)	58.7	76.7	74.1	1614.6	86.5	81.4	73.6	70.1	66.1	685	93.2 %
PDrop (CVPR25)	60.2	78.4	75.4	1643.2	87.8	82.5	75.8	70.9	67.2	690	94.6 %
HiRED (AAAI25)	58.8	76.6	74.9	1603.5	86.7	81.8	72.5	70.4	66.6	690	93.0 %
SparseVLM (ICML25)	58.8	76.7	73.9	1626.4	86.6	81.3	73.8	69.6	66.3	689	93.7 %
DivPrune (CVPR25)	<u>61.0</u>	78.7	<u>77.2</u>	1655.1	88.6	83.1	79.3	<u>71.9</u>	67.6	722	<u>96.3 %</u>
DART (EMNLP25)	60.1	80.2	76.6	1635.3	88.4	82.9	78.3	71.5	68.2	699	94.9 %
GreedyPrune (25.01)	61.2	77.8	77.0	1664.0	89.1	82.5	71.9	70.9	67.1	652	94.0 %
SPLIT (Ours)	<u>61.0</u>	<u>79.3</u>	77.5	<u>1658.3</u>	88.8	<u>83.0</u>	78.1	72.0	<u>67.9</u>	<u>720</u>	<u>96.3%</u>
Qwen2-VL-7B Token Reduction ($\downarrow 77.8\%$)											
FastV (ECCV24)	57.6	75.7	73.6	1581.0	85.4	80.1	72.0	67.3	65.2	635	90.1%
PDrop (CVPR25)	58.2	76.2	74.8	1615.5	87.2	81.4	73.3	69.9	66.7	640	91.7%
HiRED (AAAI25)	57.7	75.1	73.4	1573.9	85.9	80.8	71.8	67.6	65.9	634	89.9%
SparseVLM (ICML25)	57.6	75.5	73.3	1588.2	85.3	80.4	71.6	67.5	65.4	635	90.3%
DivPrune (CVPR25)	60.5	77.3	76.1	1621.7	88.1	82.1	76.2	<u>70.2</u>	<u>66.9</u>	657	<u>92.8%</u>
DART (EMNLP25)	58.3	78.2	75.0	1614.7	87.7	82.3	73.4	70.1	67.8	638	91.8%
GreedyPrune (25.01)	60.4	75.7	75.0	1637.2	87.8	81.1	71.6	69.3	66.1	610	91.5%
SPLIT (Ours)	60.1	<u>77.6</u>	<u>75.9</u>	<u>1636.0</u>	88.1	<u>82.1</u>	<u>74.9</u>	70.5	61.3	657	93.0%
Qwen2-VL-7B Token Reduction ($\downarrow 88.9\%$)											
FastV (ECCV24)	55.5	71.6	70.5	1433.8	81.3	79.3	61.5	68.2	63.2	420	77.6%
PDrop (CVPR25)	57.0	73.9	72.0	1493.0	84.6	81.0	65.3	69.0	64.5	484	82.1%
HiRED (AAAI25)	56.0	71.5	71.0	1434.5	81.9	79.3	61.1	68.1	63.2	401	77.0%
SparseVLM (ICML25)	55.2	71.9	70.2	1432.9	80.4	80.1	62.4	69.1	63.0	432	78.0%
DivPrune (CVPR25)	59.3	75.2	<u>73.0</u>	<u>1542.1</u>	87.9	81.5	66.7	69.1	<u>64.9</u>	<u>512</u>	<u>84.9%</u>
DART (EMNLP25)	55.1	74.5	72.3	1491.4	84.5	82.1	68.5	69.5	65.3	491	82.4%
GreedyPrune (25.01)	57.5	73.3	72.3	1509.8	85.1	80.7	65.2	68.4	64.6	502	83.2%
SPLIT (Ours)	<u>58.0</u>	75.2	73.7	1545.8	<u>86.4</u>	81.7	68.8	<u>69.4</u>	64.1	530	85.6%

Table 13. Performance comparison conducted with *Qwen2.5-VL-7B* on 10 image understanding tasks. The best score is marked in **bold**, and the second best is marked with underline.

Method	GQA	MMB	MMB-CN	MME	POPE	SQA	VQA (txt)	VQA (v2)	VizWiz	OCRBench	Avg.
Qwen2.5-VL-7B Upper Bound All Tokens (100%)											
Vanilla	60.4	82.7	80.8	1684.6	87.4	76.7	83.0	79.2	70.3	846	100%
Qwen2.5-VL-7B Token Reduction ($\downarrow$ 66.7%)											
DivPrune (CVPR25)	<u>59.0</u>	81.3	<u>78.1</u>	1681.7	86.6	76.5	<u>78.3</u>	78.9	70.4	<u>738</u>	96.1 %
DART (EMNLP25)	58.3	80.2	77.5	1676.7	85.9	74.5	75.2	77.8	68.7	659	93.1 %
GreedyPrune (25.01)	58.4	80.8	77.4	1672.6	85.1	76.4	74.6	77.2	69.8	633	92.2 %
SPLIT (Ours)	59.5	81.3	78.4	1680.9	86.2	76.5	77.9	<u>78.7</u>	<u>70.2</u>	740	96.1 %
Qwen2.5-VL-7B Token Reduction ($\downarrow$ 77.8%)											
DivPrune (CVPR25)	58.5	<u>79.3</u>	<u>76.5</u>	<u>1680.0</u>	86.0	74.0	75.5	<u>77.5</u>	69.1	<u>661</u>	93.2 %
DART (EMNLP25)	56.6	<u>78.5</u>	<u>75.4</u>	1639.3	83.4	73.6	68.8	76.5	67.6	562	88.3 %
GreedyPrune (25.01)	55.4	78.0	75.1	1605.0	81.9	<u>74.2</u>	59.8	76.2	68.0	557	86.7 %
SPLIT (Ours)	<u>57.8</u>	79.7	76.8	<u>1675.2</u>	<u>84.7</u>	74.7	<u>74.8</u>	77.8	<u>69.0</u>	666	93.2 %
Qwen2.5-VL-7B Token Reduction ($\downarrow$ 88.9%)											
DivPrune (CVPR25)	55.6	77.3	73.5	1611.2	83.2	73.2	65.4	75.2	65.8	514	85.5 %
DART (EMNLP25)	52.0	73.6	70.0	1556.7	78.6	<u>72.5</u>	55.8	<u>75.5</u>	65.1	420	80.0 %
GreedyPrune (25.01)	51.6	70.1	65.7	1545.9	76.5	69.4	55.7	74.8	64.2	427	79.4 %
SPLIT (Ours)	<u>54.7</u>	<u>76.6</u>	<u>72.9</u>	<u>1608.7</u>	<u>83.1</u>	73.9	<u>64.7</u>	76.1	66.3	520	85.6 %

 Table 14. Experiments on *Qwen2.5-VL-32B*.

Method	GQA	MMB	MMB-CN	MME	POPE	SQA	VQA ^{txt}	Avg.
Qwen2.5-VL-32B Upper Bound, All Tokens (100%)								
Vanilla	59.2	86.3	83.7	1732.5	86.1	67.2	77.1	100.0%
Qwen2.5-VL-32B Token Reduction ($\downarrow$ 66.7%)								
DivPrune (CVPR25)	<u>58.0</u>	<u>84.5</u>	<u>81.1</u>	<u>1693.2</u>	<u>84.5</u>	<u>61.7</u>	<u>73.5</u>	97.5%
DART (EMNLP25)	56.8	83.4	80.3	1655.6	84.0	61.6	72.8	95.5%
GreedyPrune (25.01)	57.0	83.9	80.5	1686.3	83.7	60.6	72.8	96.9%
SPLIT (Ours)	58.1	84.6	81.4	1695.9	84.6	62.1	74.0	97.7%
Qwen2.5-VL-32B Token Reduction ($\downarrow$ 77.8%)								
DivPrune (CVPR25)	57.1	83.0	79.2	1667.4	83.4	58.7	70.6	95.8%
DART (EMNLP25)	54.8	80.2	76.9	1622.8	80.4	59.5	68.2	93.2%
GreedyPrune (25.01)	54.3	80.0	81.8	1616.4	81.0	55.7	61.9	92.7%
SPLIT (Ours)	<u>56.1</u>	<u>82.0</u>	<u>80.8</u>	<u>1663.1</u>	<u>82.3</u>	<u>59.0</u>	<u>69.7</u>	95.5%
Qwen2.5-VL-32B Token Reduction ($\downarrow$ 88.9%)								
DivPrune (CVPR25)	53.5	<u>78.1</u>	<u>75.1</u>	1564.5	79.4	53.2	<u>62.1</u>	89.7%
DART (EMNLP25)	49.1	73.7	70.7	1505.1	72.2	54.5	56.2	85.8%
GreedyPrune (25.01)	49.8	73.2	75.0	1495.1	74.7	53.9	55.8	85.6%
SPLIT (Ours)	<u>53.1</u>	78.7	75.2	<u>1563.5</u>	<u>77.2</u>	<u>55.2</u>	63.7	89.7%

 Table 15. Experiments on *Qwen2.5-VL-72B*.

Method	GQA	MMB	MMB-CN	MME	POPE	SQA	VQA ^{txt}	Avg.
Qwen2.5-VL-72B Upper Bound, All Tokens (100%)								
Vanilla	58.2	88.5	86.5	1737.9	86.4	77.4	81.9	100.0%
Qwen2.5-VL-72B Token Reduction ($\downarrow$ 66.7%)								
DivPrune (CVPR25)	56.9	86.6	85.2	1729.2	<u>85.4</u>	75.9	78.5	99.1%
DART (EMNLP25)	<u>56.8</u>	<u>86.7</u>	85.4	<u>1737.4</u>	85.7	76.3	79.9	99.6%
GreedyPrune (25.01)	53.4	85.3	84.2	1700.9	85.2	<u>76.1</u>	<u>78.9</u>	97.9%
SPLIT (Ours)	56.3	86.8	85.4	1735.4	85.2	<u>76.1</u>	<u>78.9</u>	99.4%
Qwen2.5-VL-72B Token Reduction ($\downarrow$ 77.8%)								
DivPrune (CVPR25)	56.3	84.7	84.0	1683.5	<u>84.3</u>	74.5	75.9	96.7%
DART (EMNLP25)	55.2	85.5	84.0	<u>1700.9</u>	85.2	76.2	77.4	97.7%
GreedyPrune (25.01)	51.2	83.8	82.4	1625.4	75.2	73.1	74.4	93.2%
SPLIT (Ours)	<u>55.8</u>	<u>85.2</u>	84.0	1705.9	84.4	<u>75.0</u>	<u>76.7</u>	97.7%
Qwen2.5-VL-72B Token Reduction ($\downarrow$ 88.9%)								
DivPrune (CVPR25)	53.7	82.2	82.1	1587.4	82.4	72.6	67.9	91.5%
DART (EMNLP25)	51.4	81.0	81.5	<u>1605.5</u>	79.8	74.7	70.5	92.2%
GreedyPrune (25.01)	48.3	79.3	78.6	1589.4	74.0	72.0	65.5	90.5%
SPLIT (Ours)	<u>53.3</u>	<u>82.0</u>	<u>81.8</u>	1606.4	<u>81.3</u>	<u>74.0</u>	<u>70.4</u>	92.4%

B.2. Comparison across layer selections

As briefly discussed in the main text (Figure 6a), applying temporal shift across all layers gives the best overall performance, while middle layers are the most effective among single groups. Here, we further compare different ways of selecting mixed layer subsets. As shown in Table 16, most mixed configurations achieve very similar results, indicating that the method is robust to the exact choice of layers. However, when the lowest layers (*e.g.*, layer 0) are included, performance drops noticeably, suggesting that shallow representations are less informative for temporal shift. In contrast, combinations that involve only middle and higher layers remain consistently strong across different token budgets, highlighting the complementary role of representations from deeper layers. These findings further confirm the generality of our approach across layer selection strategies.

 Table 16. Results of different Mixed layer selections on *LLaVA-Next-7B*.

Method	GQA	MMB	MMB-CN	MME	POPE	SQA	VQA ^{Text}	Avg.
LLaVA-Next-7B Upper Bound, All Tokens (100%)								
Vanilla	62.2	66.3	57.0	1483.5	87.7	69.3	69.3	100.0%
Token Reduction ($\downarrow$ 66.7%)								
Mixed [0,1,8,9,16,17]	60.5	65.0	55.0	1465.0	86.8	68.7	61.0	98.9%
Mixed [1,2,9,10,17,18]	61.8	66.2	56.1	1480.1	87.5	69.7	62.5	99.7%
Mixed [2,3,10,11,19,20]	61.7	66.1	56.2	1479.0	87.4	69.9	62.6	99.6%
Mixed [3,4,11,12,20,21]	61.6	66.0	56.1	1478.6	87.4	69.8	62.7	99.6%
Mixed [4,5,12,13,21,22]	61.5	65.9	56.0	1477.9	87.3	69.7	62.4	99.5%
Token Reduction ($\downarrow$ 77.8%)								
Mixed [0,1,8,9,16,17]	60.0	64.2	53.9	1452.0	86.2	67.5	59.5	97.9%
Mixed [1,2,9,10,17,18]	61.3	65.3	54.9	1466.1	87.1	68.4	60.9	98.7%
Mixed [2,3,10,11,19,20]	61.4	65.4	55.0	1467.0	87.0	68.5	60.8	98.8%
Mixed [3,4,11,12,20,21]	61.2	65.2	54.9	1465.2	87.0	68.4	60.7	98.6%
Mixed [4,5,12,13,21,22]	61.1	65.1	54.8	1464.7	86.9	68.3	60.6	98.6%
Token Reduction ($\downarrow$ 88.9%)								
Mixed [0,1,8,9,16,17]	59.0	64.0	52.7	1398.0	85.5	67.5	54.0	95.0%
Mixed [1,2,9,10,17,18]	60.4	65.3	53.7	1412.0	86.3	68.6	55.4	96.1%
Mixed [2,3,10,11,19,20]	60.5	65.4	53.8	1413.0	86.2	68.7	55.5	96.2%
Mixed [3,4,11,12,20,21]	60.3	65.2	53.9	1411.2	86.2	68.6	55.4	96.1%
Mixed [4,5,12,13,21,22]	60.3	65.2	53.8	1411.0	86.1	68.6	55.3	96.1%

C. Visual Analysis

C.1. Detailed Analysis of Pivot

As discussed in the main text, diversity-based pruning methods often suffer from redundant pivot selections, since high-norm tokens tend to be similar to one another. To further illustrate this issue, we provide additional qualitative examples in Figure 7. These cases show that pivot tokens with large norms are frequently clustered in neighboring regions, leading to overlapping or nearly identical token selections. Such redundancy reduces the diversity of preserved tokens and results in poor coverage of other important areas. Interestingly, the opposite phenomenon can also occur, where low-norm but dissimilar tokens are excluded, leaving entire regions underrepresented. The supplementary examples in Figure 7 thus reinforce our claim that naive pivot-based strategies may lead to both over-sampling and under-sampling, highlighting the need for more robust criteria that consider token similarity in addition to magnitude.

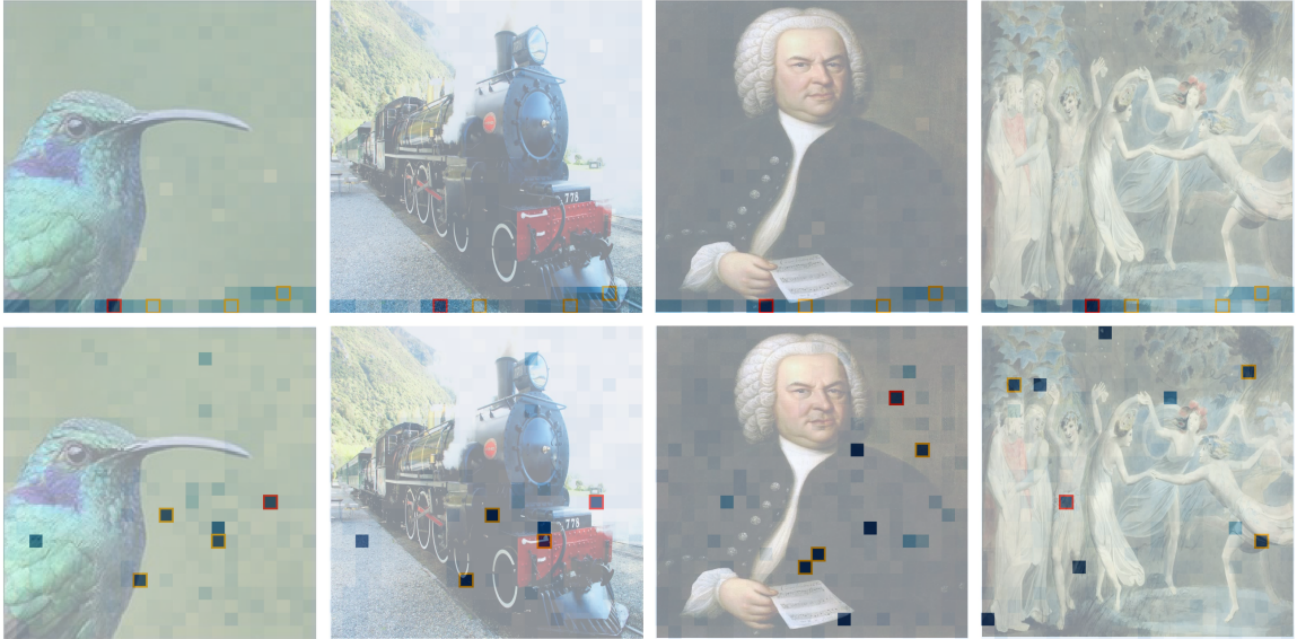


Figure 7. Limitations of pivot tokens. The red box marks the first pivot, with the heatmap showing its cosine similarity to other tokens. Subsequent pivots (orange boxes) largely overlap with it, revealing redundancy in the pivot selection process (top: pivots chosen from high-norm tokens; bottom: the opposite case).

Figure 8 shows that increasing the number of pivots results in only marginal changes to the selected token set. Even with multiple pivots, the selected tokens largely coincide, suggesting that the selection is dominated by a small subset of similar tokens. This indicates that simply adding more pivots does not effectively improve coverage or diversity.

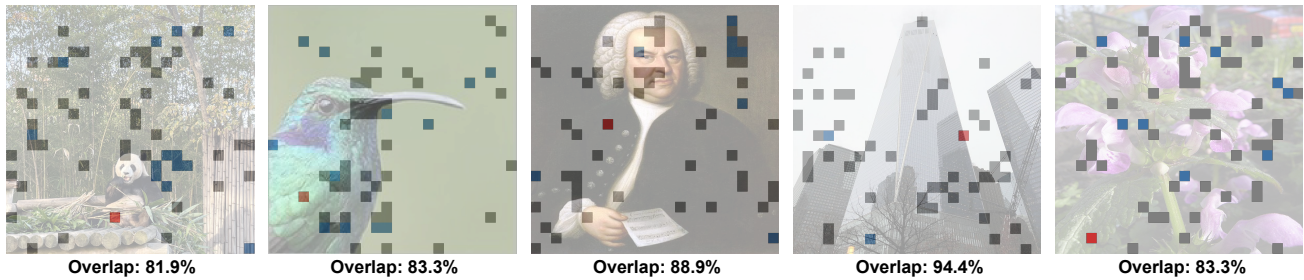


Figure 8. Pivot selection shows clear limitations. The red box marks a single pivot, blue boxes are tokens added only when multiple pivots are used, and gray boxes show overlaps. The high overlap reveals that single-pivot selection already covers most of the multi-pivot result, limiting diversity.

C.2. Visualization Results

Figure 9 presents visualization results comparing *SPLIT* with prior pruning methods. FastV exhibits strong position bias, with most tokens concentrated in the lower region (*Lim. 2*). HiRED relies solely on token importance, leading to low diversity and missing background cues, which limits global understanding. Diversity-based approaches mitigate this bias but often allocate pivots to a single region, resulting in poor coverage (*Lim. 3*). In contrast, *SPLIT* maintains both importance and diversity, producing balanced pivots and more faithful visual coverage.

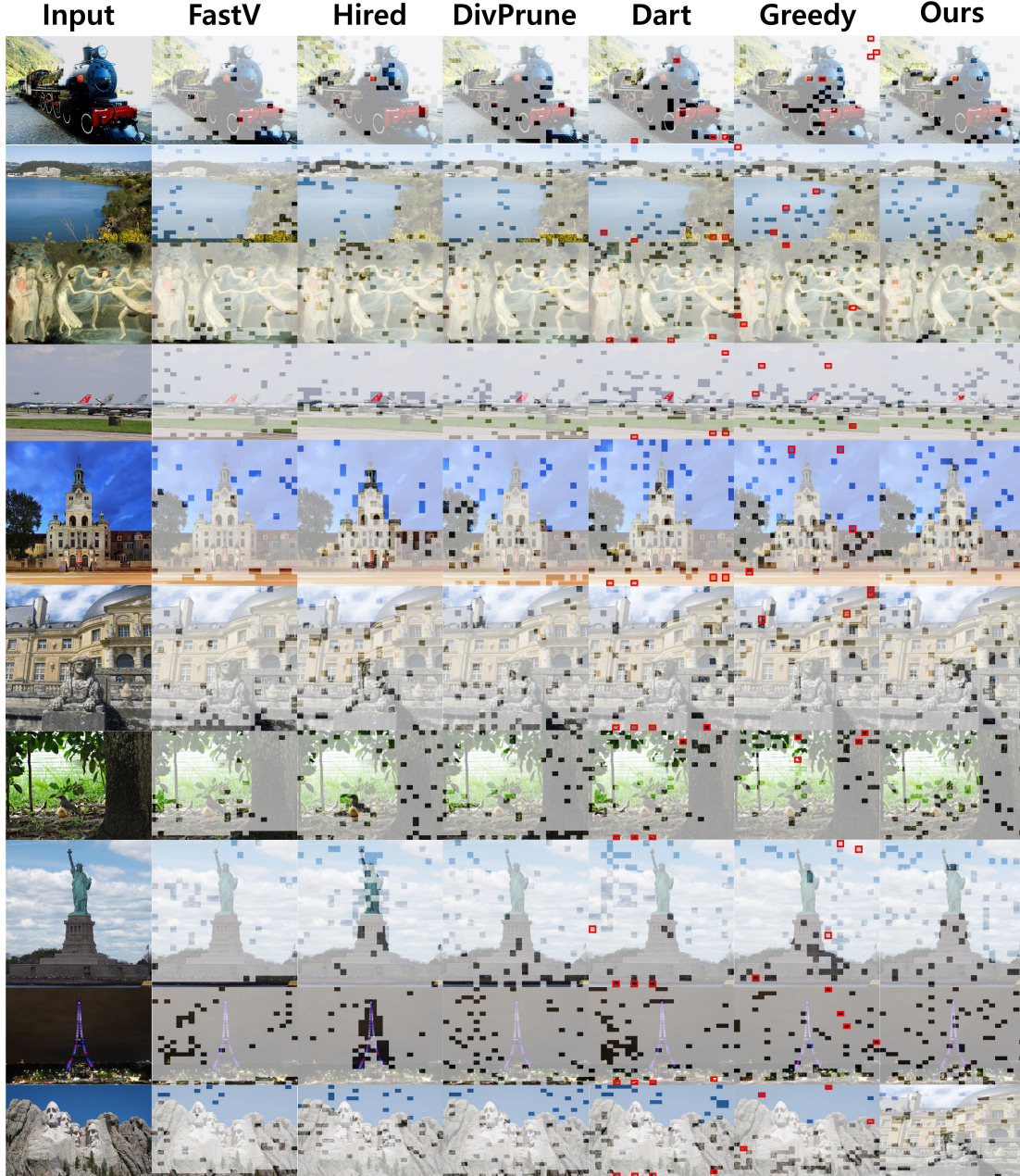


Figure 9. Visualization of token dropping results across different methods. Red tokens denote pivots.